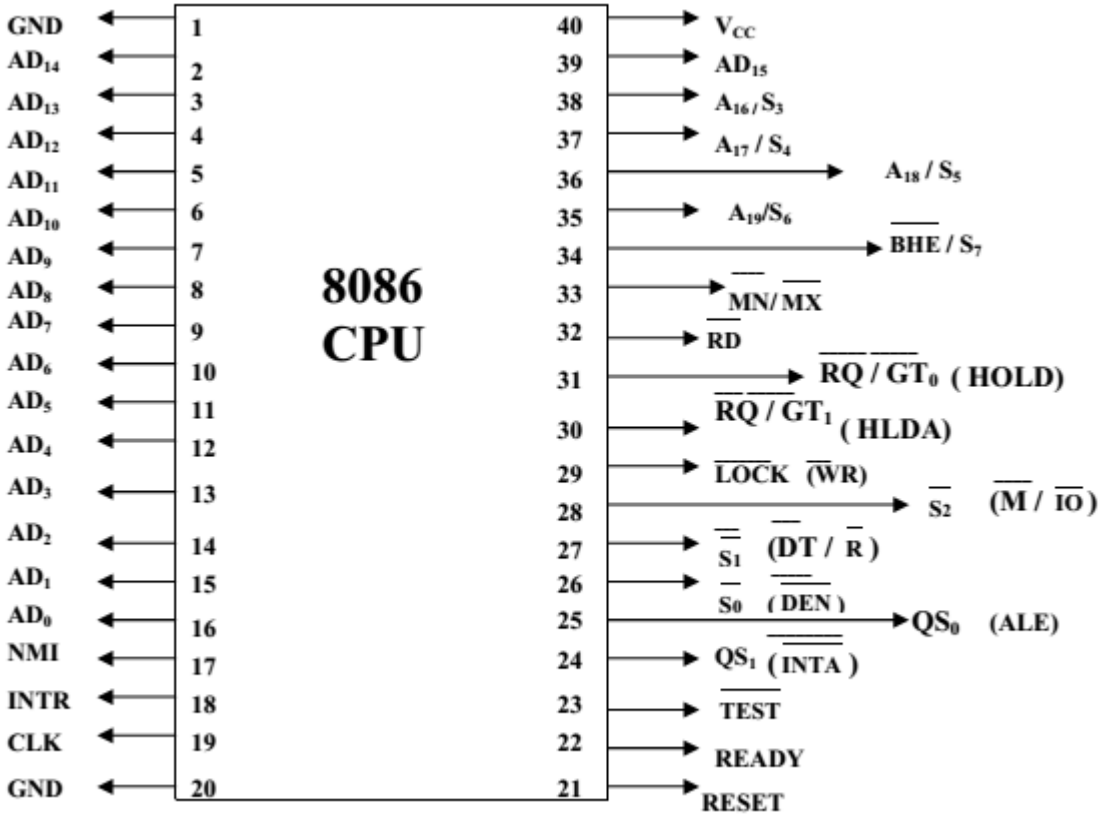
# EEM 364
# Mikroişlemcileri

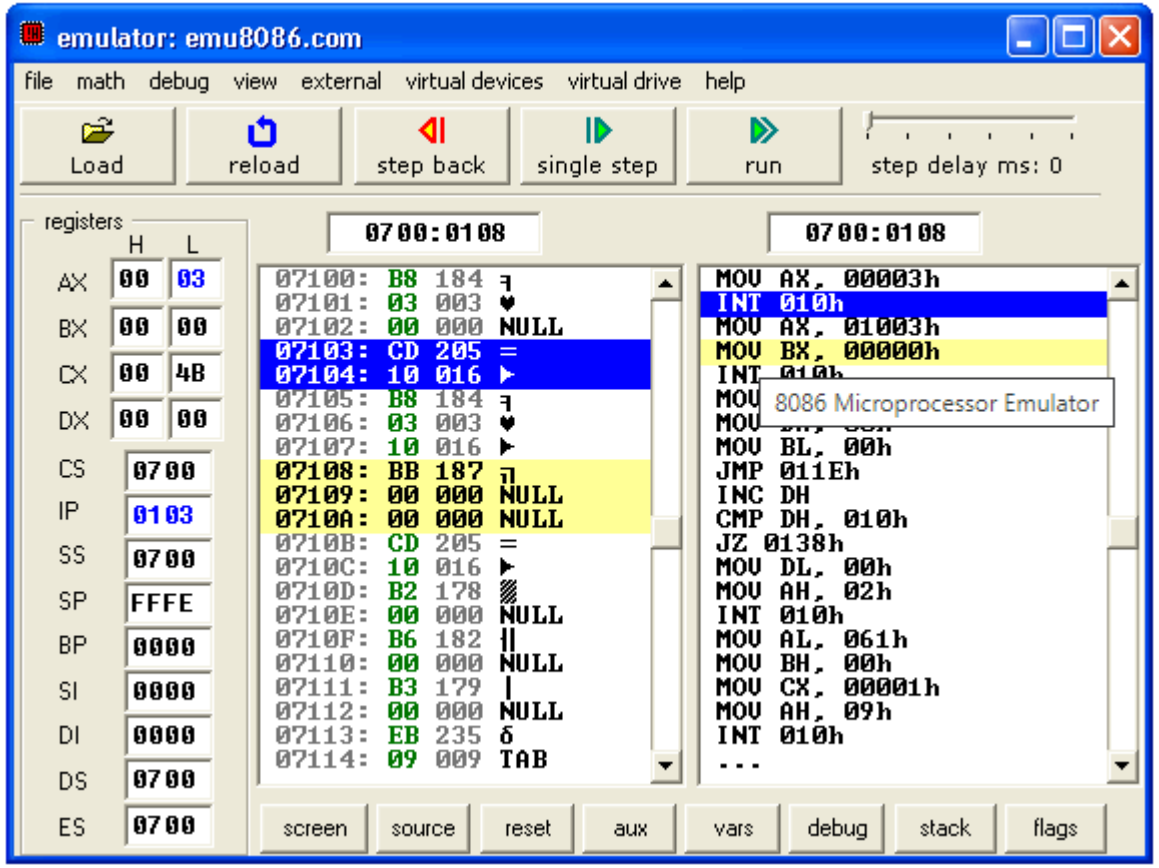
# Kaynaklar


- https://www.youtube.com/watch?v=CLB57X3NfjE&list=PLnuIwzdkcskIn5LtQFAKFUET25EAUu4DL&index=1&ab_channel=Ba%C5%9FkentAkademi
- https://www.youtube.com/watch?v=CYWIrXjXDkI&list=PLojMMqrzaR2nQr9513hT3F8gkrJVUrKwQ&ab_channel=%C5%9Euayb%C3%87a%C4%9Fr%C4%B1Yener
- https://www.youtube.com/watch?v=BPFvtfZmkH4&ab_channel=Mert%C3%96ZEN
- https://web.itu.edu.tr/buzluca/cse342/notes.html
- https://web.karabuk.edu.tr/emelkocak/indir/MTM305/1x1.pdf
- https://faraday.emu.edu.tr/eeng410/lectures/eeng410-Lecture1.pdf
- http://tuncayuzun.com/DersNot_MikSis.htm

# Anlatılacak Olan uP



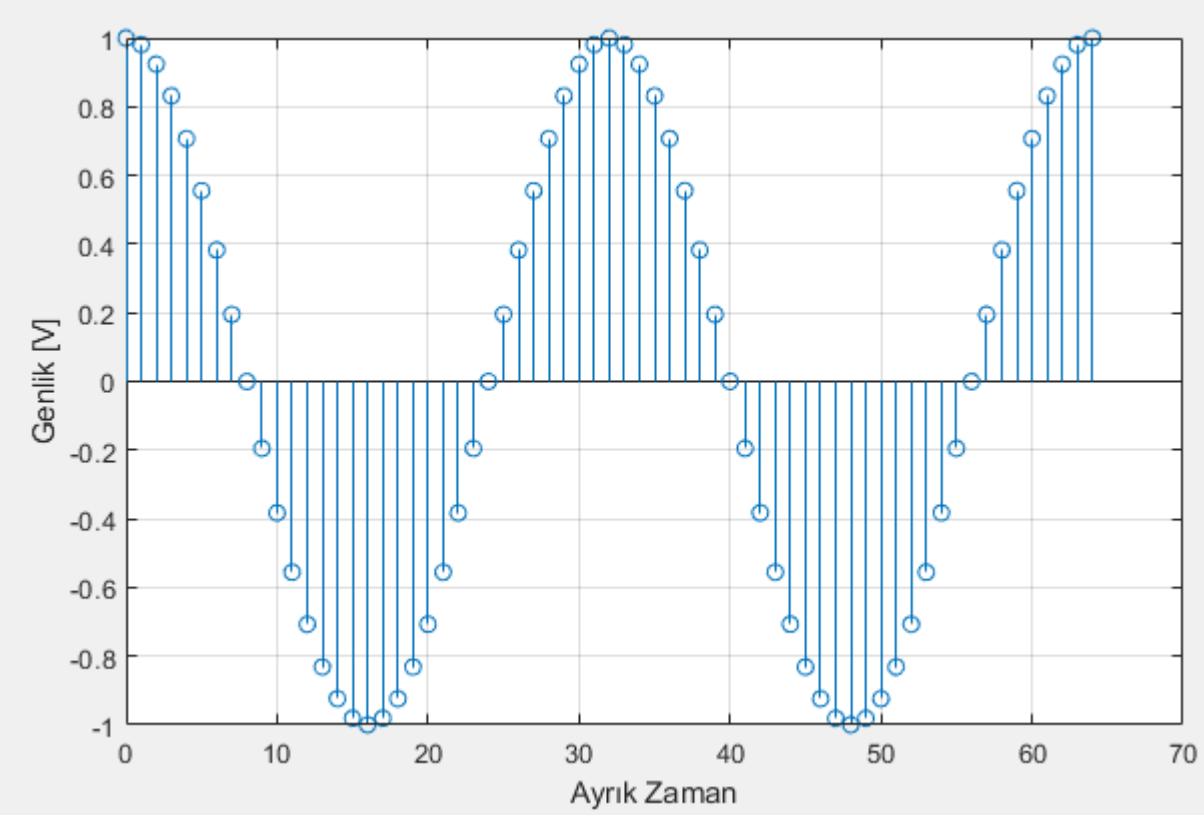

Pin Diagram of 8086



# Mikroişlemciler Dersine Giriş

## Temel Kavramlar

# Mikroişlemcileri: Elektronik Devre Temelleri

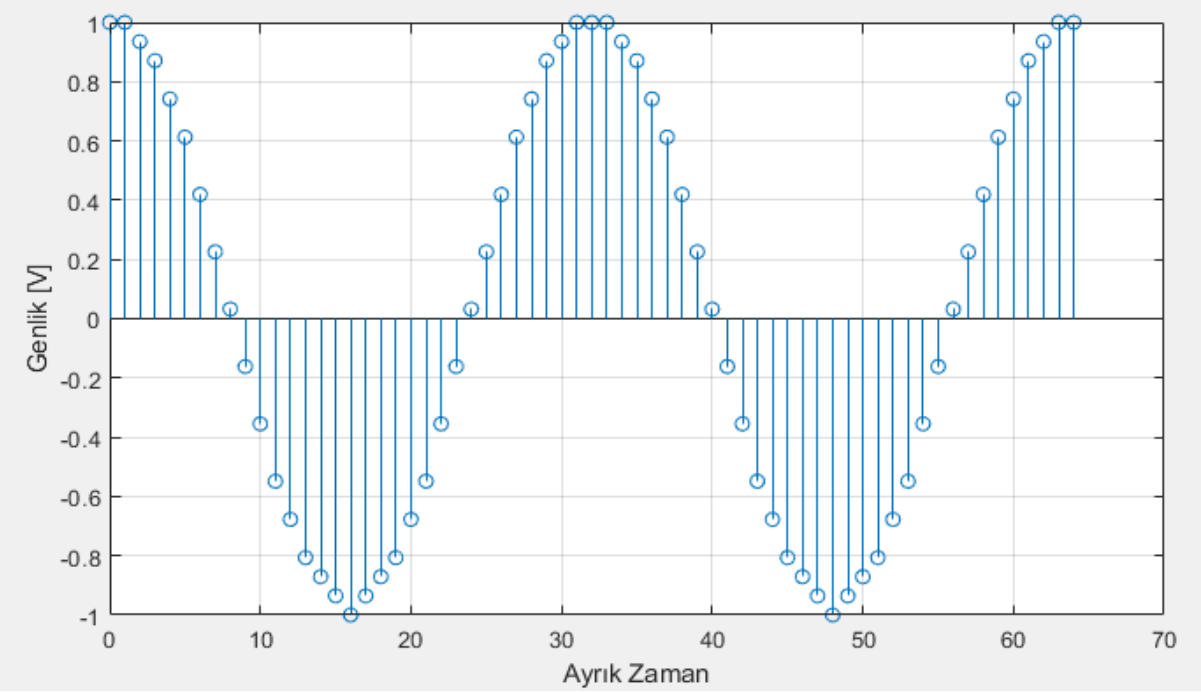



**Başlangıç:** **Bilginin elektrik sinyaline dönüşmesi**

**Ama analog işaretleri işlemesi zor !!**

→ **Örnekleme**

# Mikroişlemcileri: Elektronik Devre Temelleri




**Başlangıç: Bilginin elektriksel sinyaline dönüşmesi**

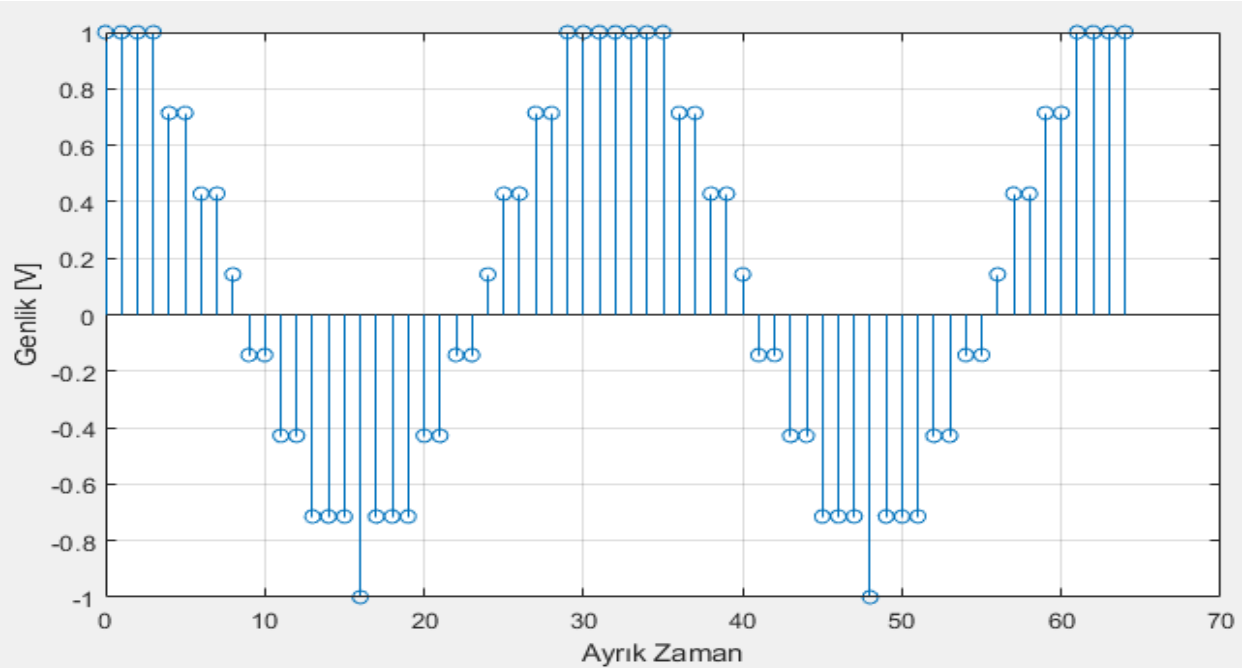
**Ama analog işaretleri işlemesi zor !!**

→ **Örnekleme**

→ **Zamanda Örnekleme**

→ **Genlikte Örnekleme (Dijital İşaret)**

# Mikroişlemcileri: Elektronik Devre Temelleri




**Başlangıç:** **Bilginin elektriksel sinyaline dönüşmesi**

**Ama analog işaretleri işlemesi zor !!**

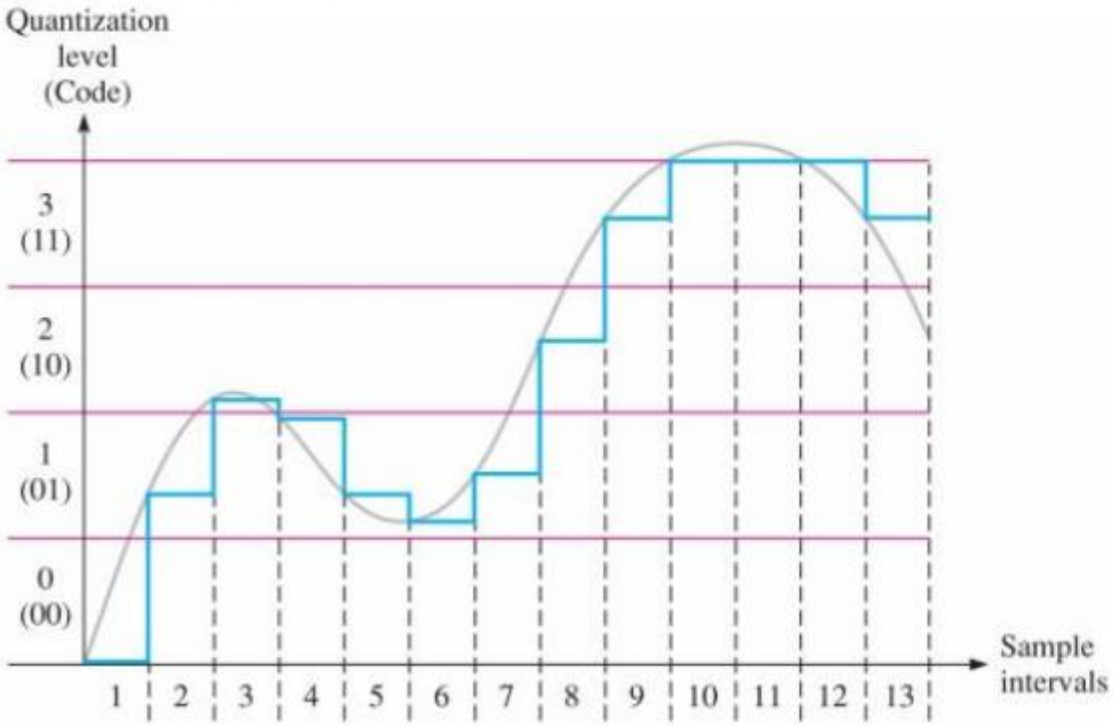
→ **Örnekleme**

→ **Zamanda Örnekleme**

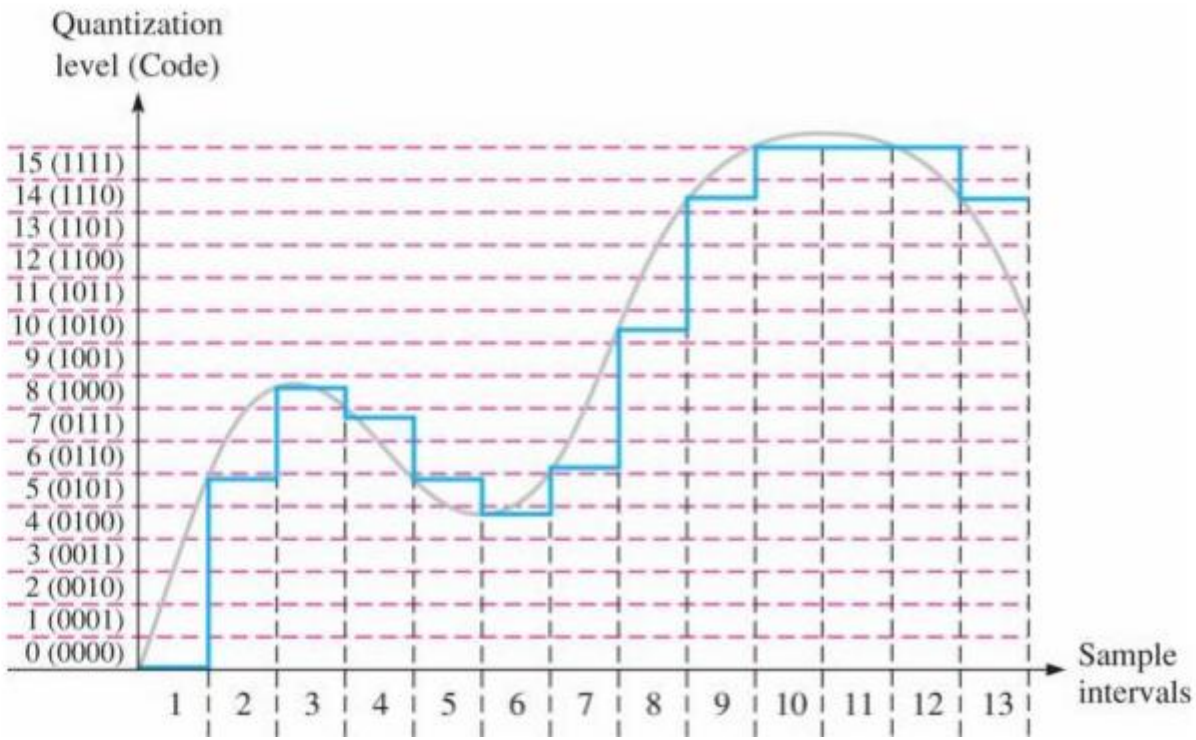
→ **Genlikte Örnekleme (Dijital İşaret)**

# Mikroişlemcileri: Elektronik Devre Temelleri

Kodlama, kuantalama düzeylerine ikili sayı sisteminde birer kod verme işlemidir.




4-bit




https://avesis.yildiz.edu.tr/ayten/dokumanlar

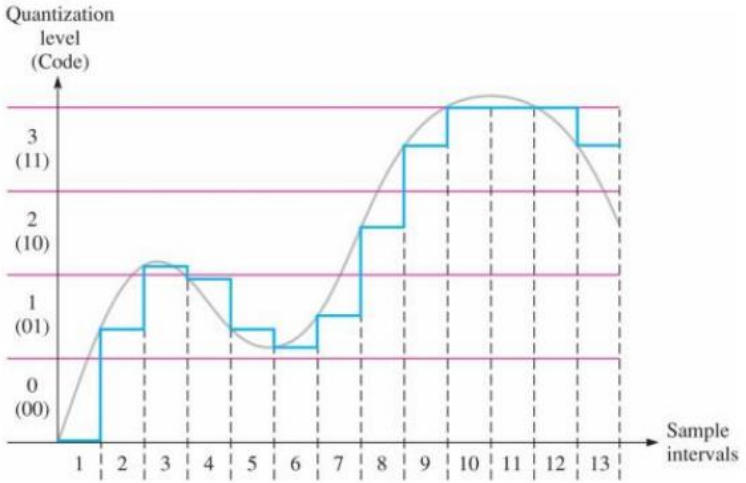
# Mikroişlemcileri: Elektronik Devre Temelleri

→ İşlemek

→ İletmek

→ Saklamak

Kodlama, kuantalama düzeylerine ikili sayı sisteminde birer kod verme işlemidir.

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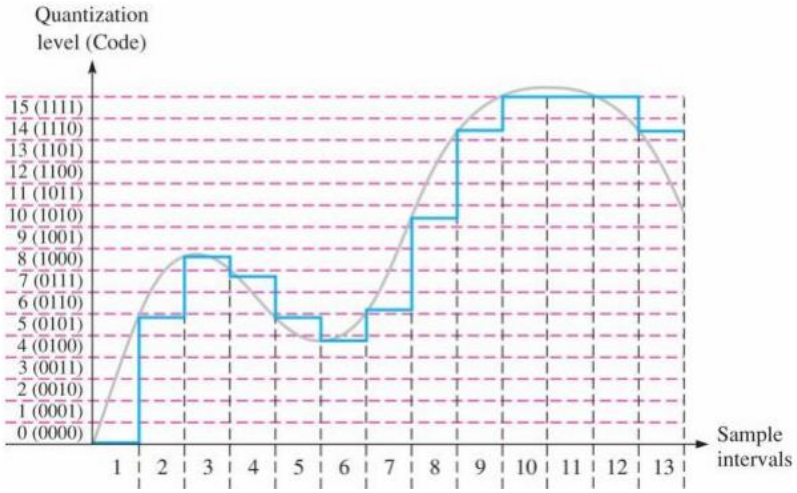
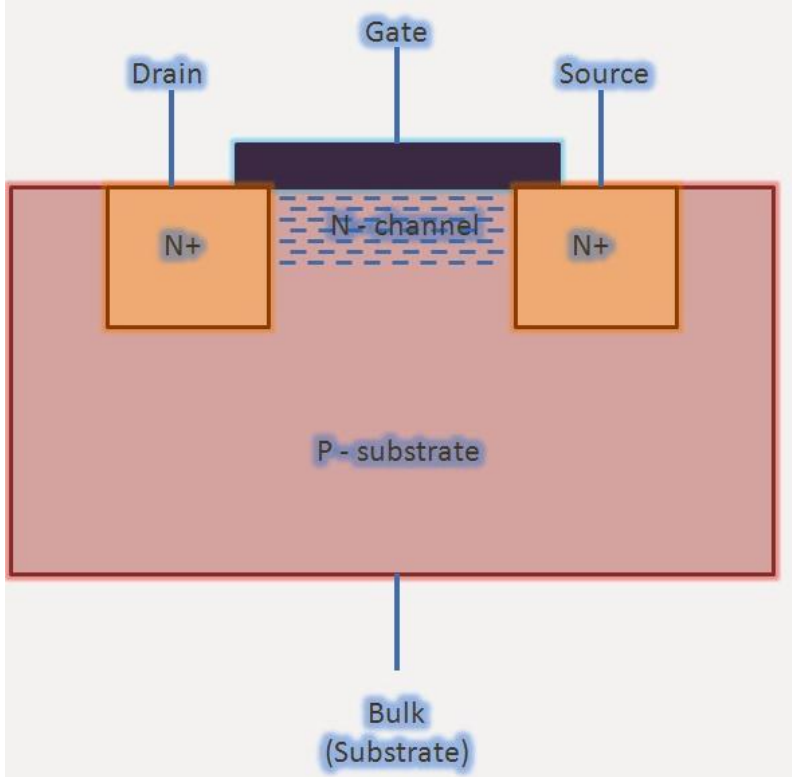


# Mikroişlemcileri: Elektronik Devre Temelleri

*Bitler Nasıl Oluşur?*




https://vlsiuniverse.blogspot.com/2013/10/regions-of-operation-of-mos-transistors.html

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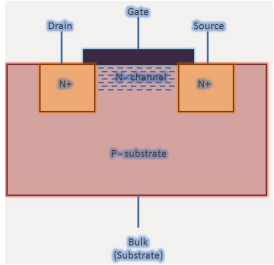
# Mikroişlemcileri: Elektronik Devre Temelleri

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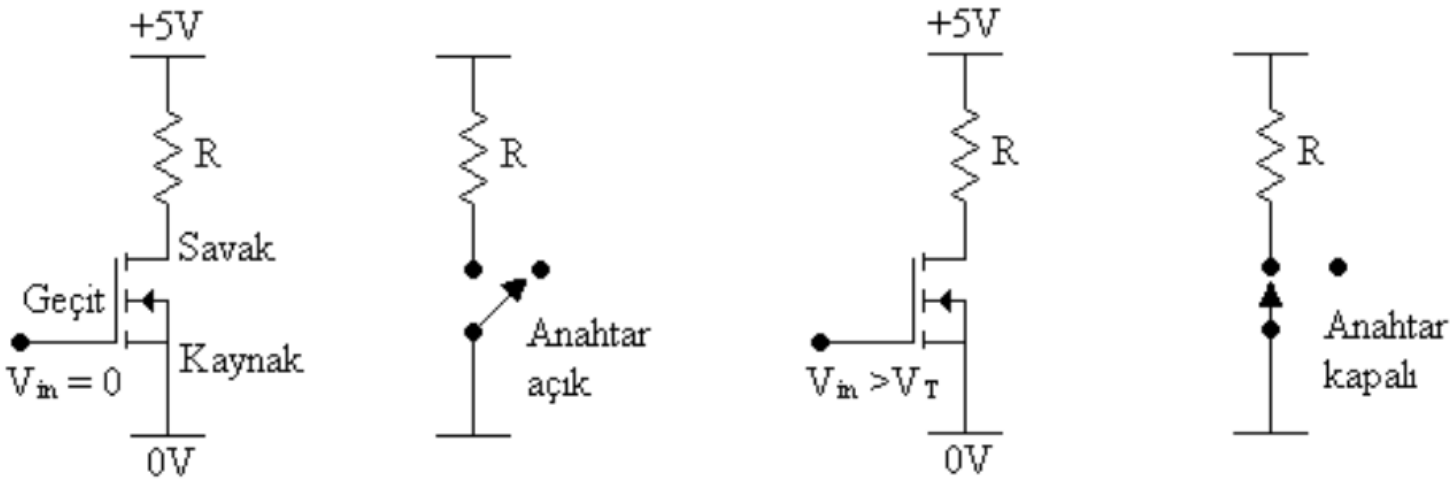
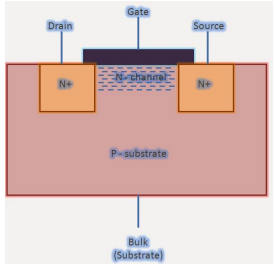


| $V_{Giriş}$ | $V_{Çıkış}$ |
|---|---|
| 0 | 1 |
| 1 | 0 |

https://vlsiuniverse.blogspot.com/2013/10/regions-of-operation-of-mos-transistors.html

https://avesis.yildiz.edu.tr/ayten/dokumanlar

# Mikroişlemcileri: Elektronik Devre Temelleri

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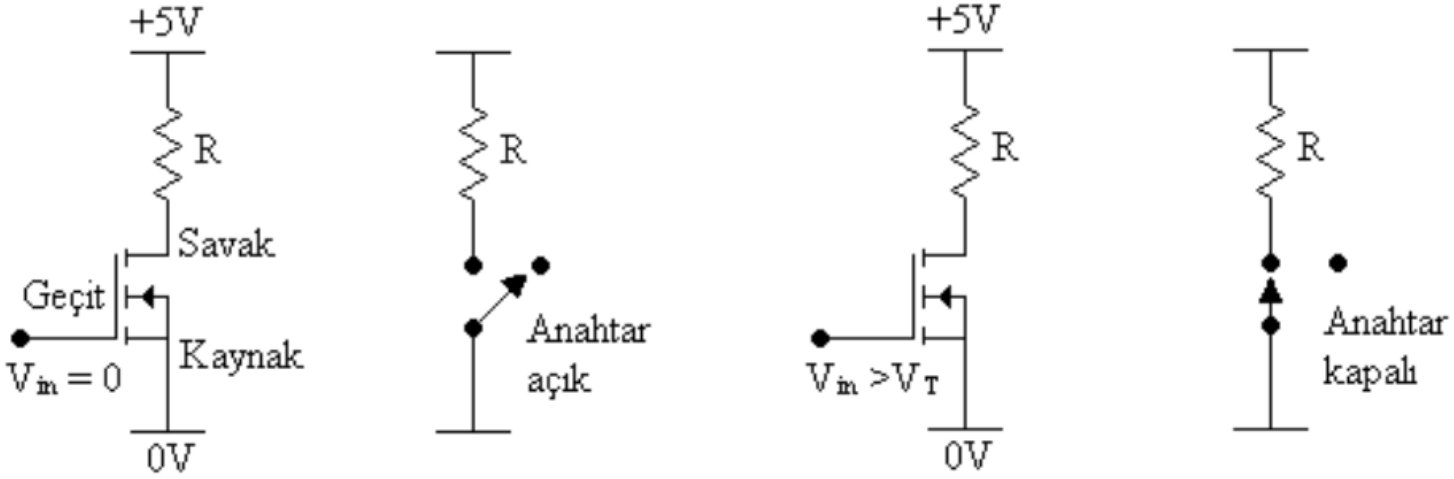

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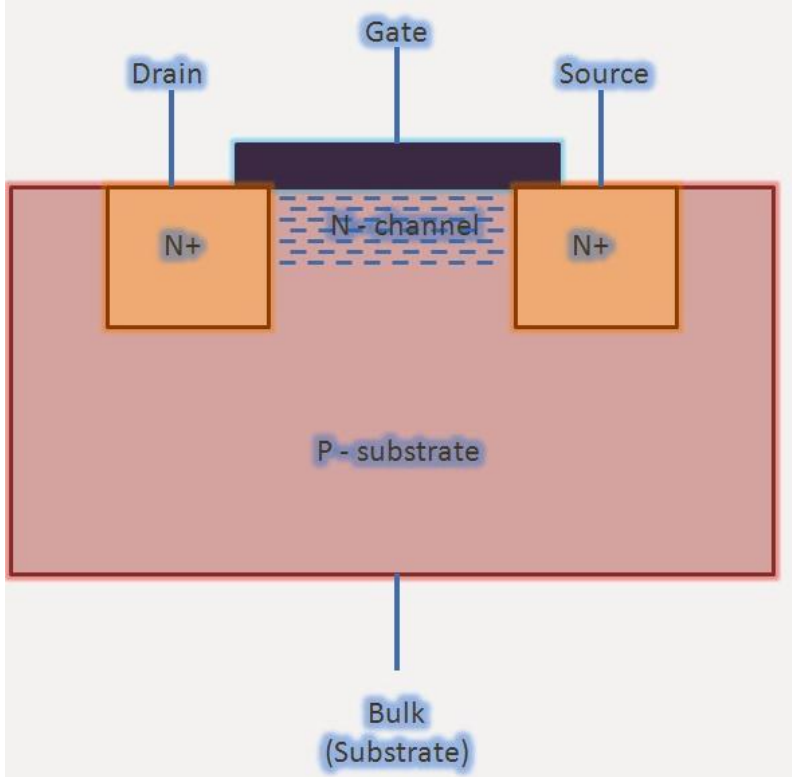
https://avesis.yildiz.edu.tr/ayten/dokumanlar

# Resistor Transistor Level

# Mikroişlemcileri: Elektronik Devre Temelleri

*Bitler Nasıl Oluşur?*

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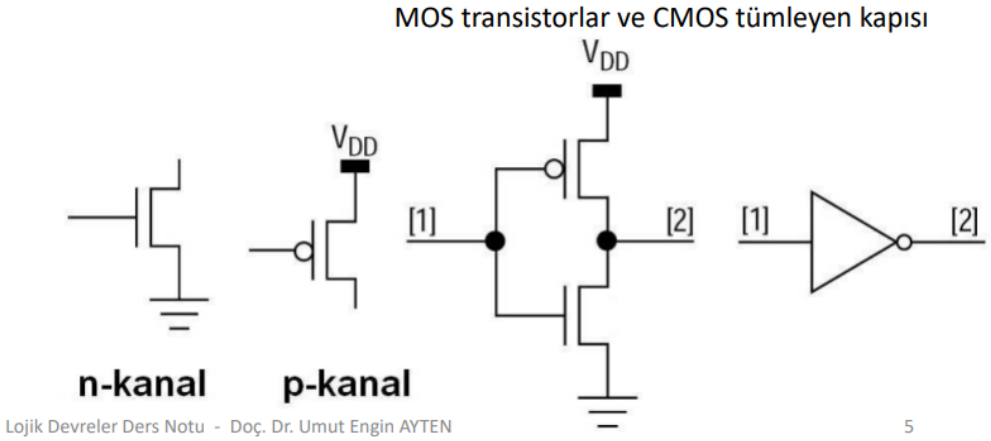


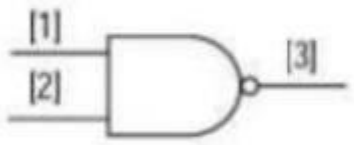
| $V_{Giriş}$ | $V_{Çıkış}$ |
|---|---|
| 0 | 1 |
| 1 | 0 |

https://avesis.yildiz.edu.tr/ayten/dokumanlar

# Mikroişlemcileri: Elektronik Devre Temelleri

*Kapılar Nasıl Oluşur?*

## VEDEĞİL (NAND) Kapısı




| Girişler | | Çıkış |
|---|---|---|
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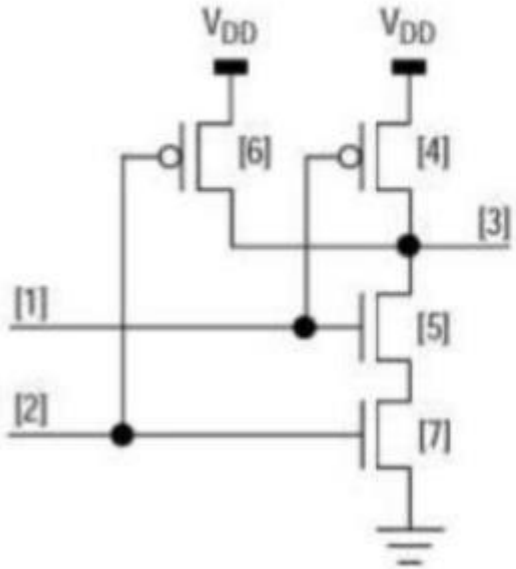
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |




CMOS VE DEĞİL Kapısı ve iç devresi

# Mikroişlemcileri: Elektronik Devre Temelleri

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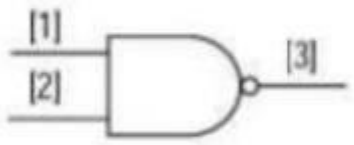
**VEDEĞİL (NAND) Kapısı**

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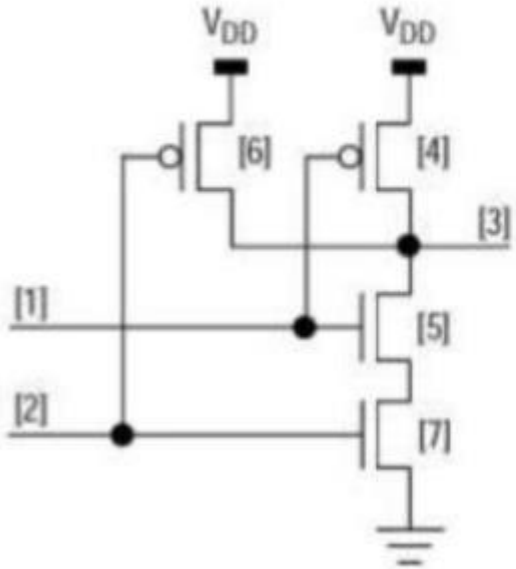


CMOS VE DEĞİL Kapısı ve iç devresi

# AND kapısı nasıl üretilir?

# Mikroişlemcileri: Elektronik Devre Temelleri

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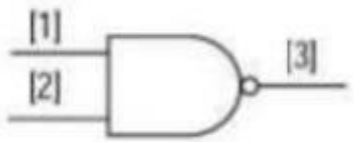
**VEDEĞİL (NAND) Kapısı**




| Girişler | | Çıkış |
|---|---|---|
| [1] | [2] | [3] |
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| 0 | 1 | 1 |
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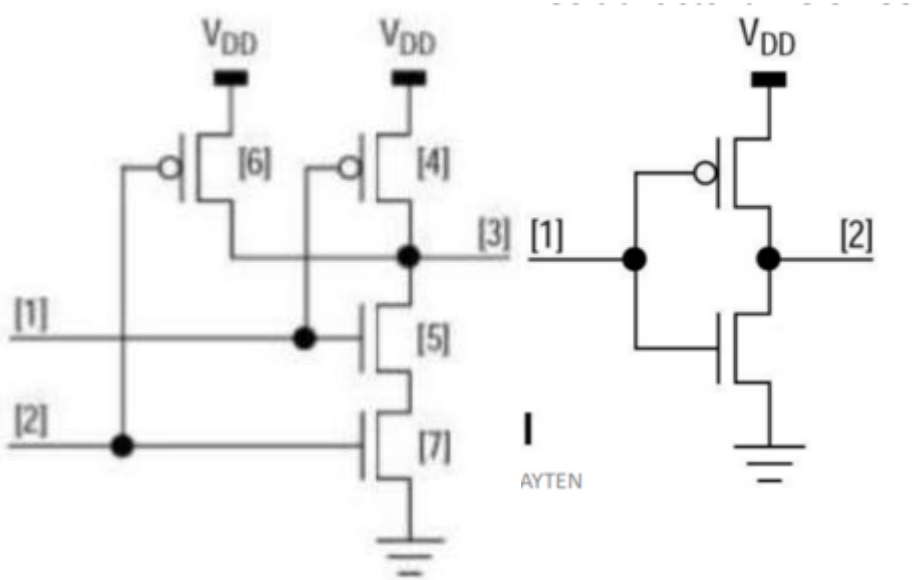
| 1 | 1 | 0 |




İL Kapısı ve iç devresi

https://avesis.yildiz.edu.tr/ayten/dokumanlar

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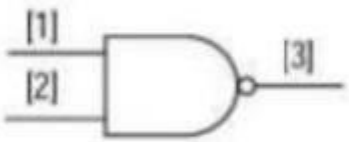
# Mikroişlemcileri: Elektronik Devre Temelleri

*Kapılar Nasıl Oluşur?*

**VEDEĞİL (NAND) Kapısı**




| Girişler | | Çıkış |
|---|---|---|
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| 0 | 1 | 1 |
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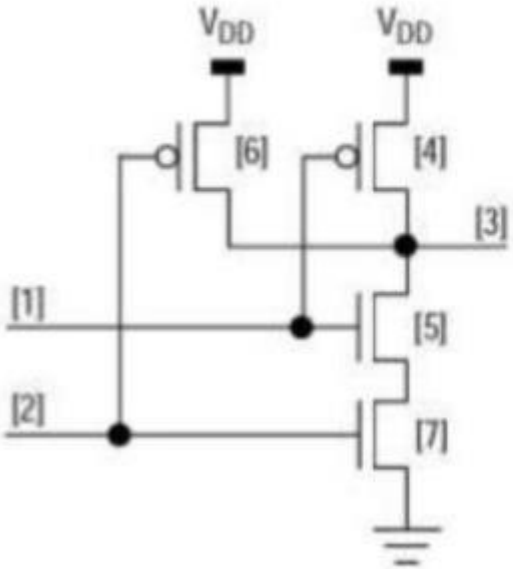


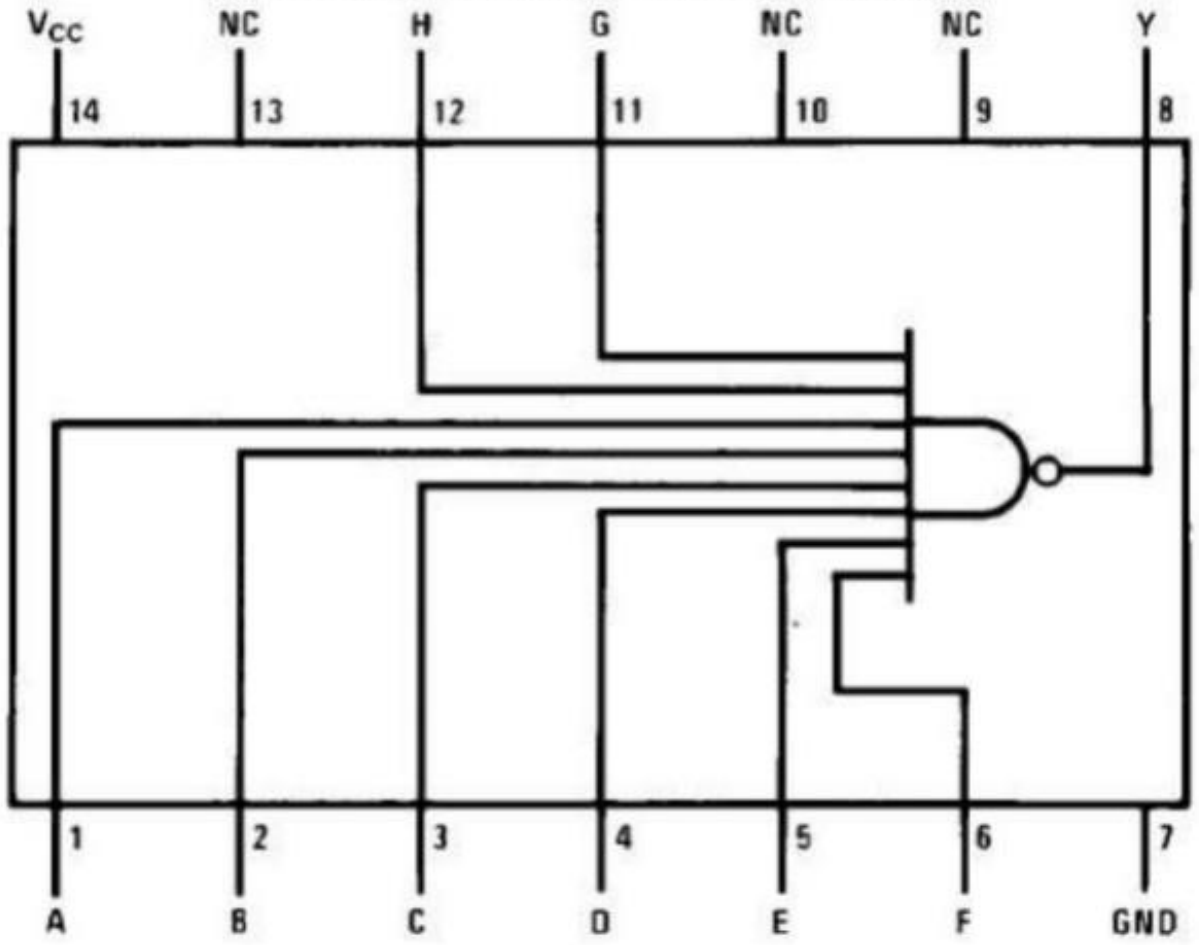

CMOS VE DEĞİL Kapısı ve iç devresi

https://avesis.yildiz.edu.tr/ayten/dokumanlar

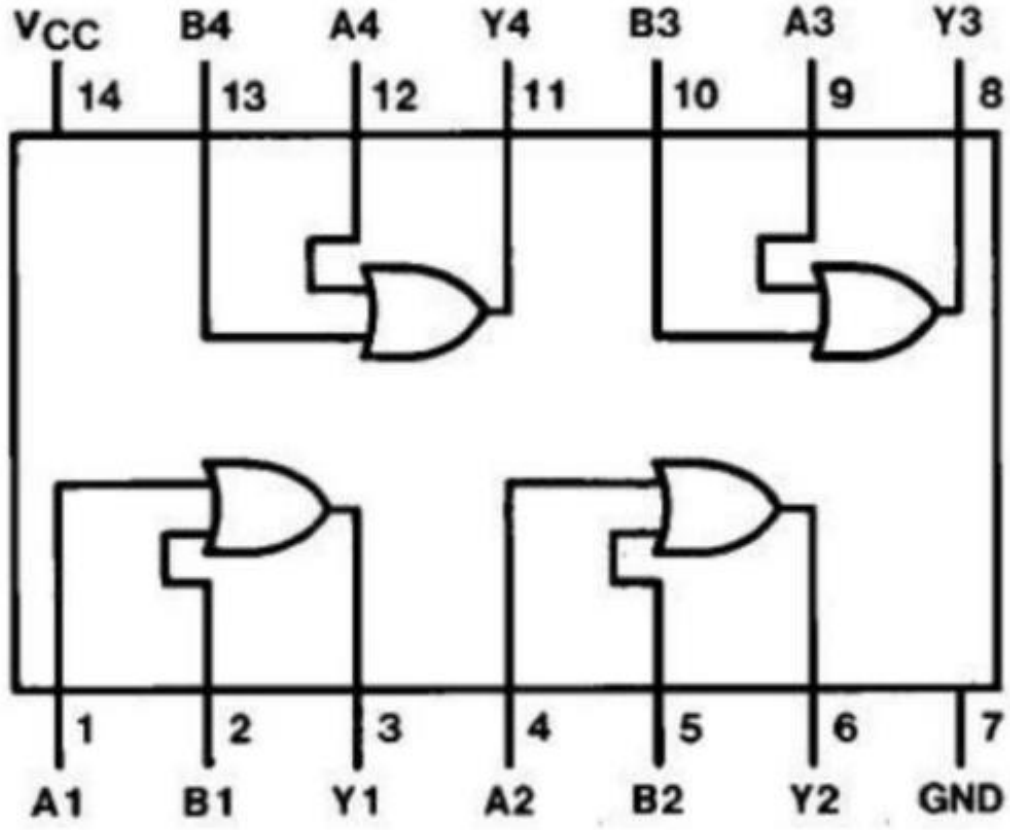
# Transistor Transistor Level

# LOJİK KAPILAR ve TÜMLEŞİK DEVRE OLARAK GERÇEKLEMELERİ

**74LS30, TTL 8-Girişli VE DEĞİL Kapısı:**

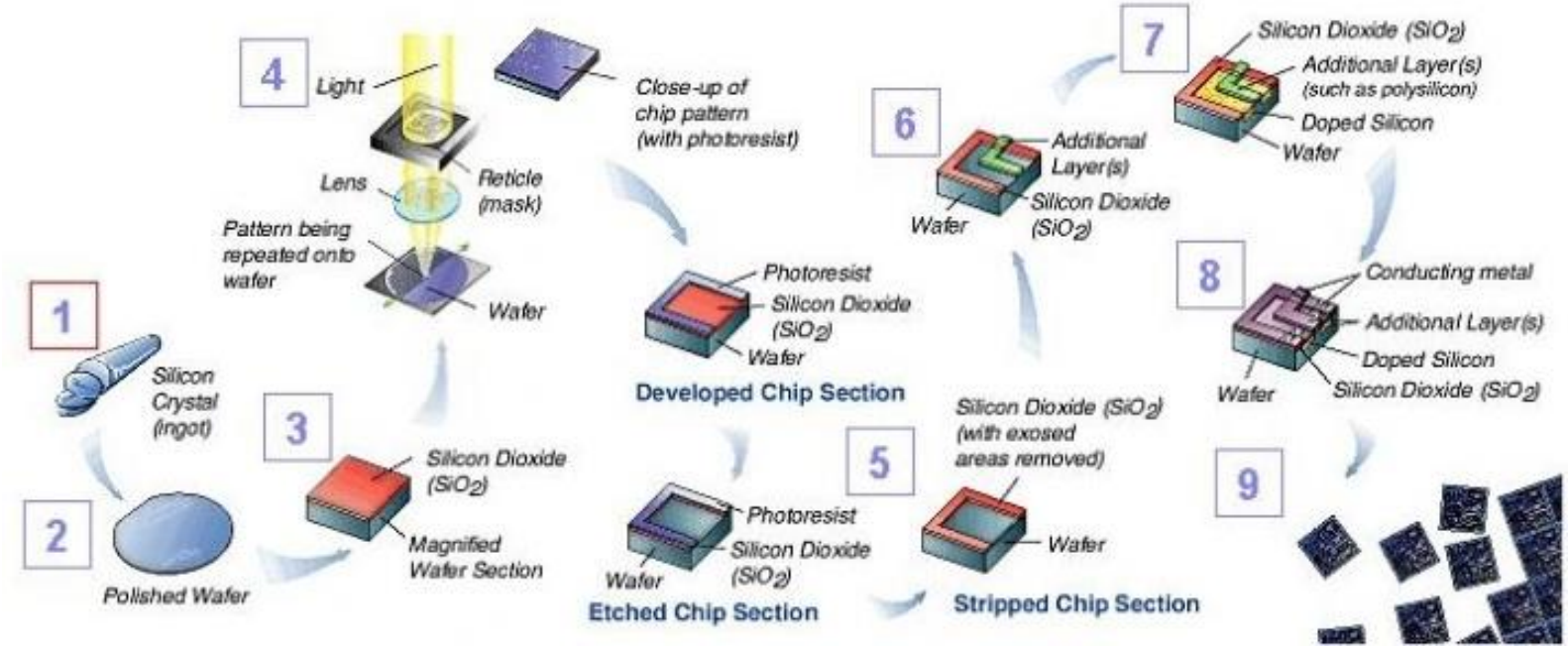


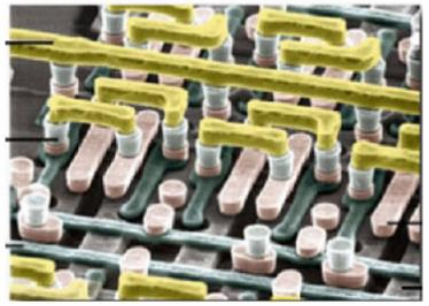

**74LS32, TTL 2-Girişli Dörtlü VEYA Kapısı:**

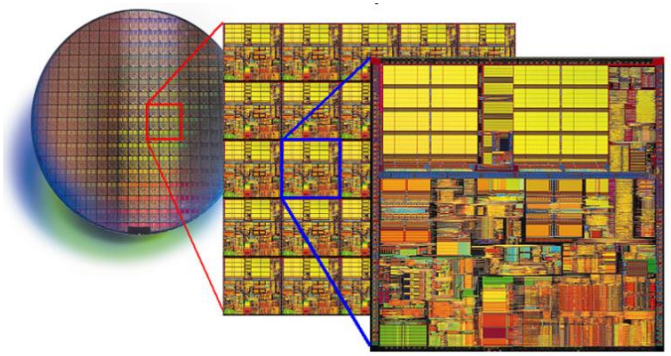


# Mikroişlemcileri: Elektronik Devre Temelleri

*Yonga Üretimi*



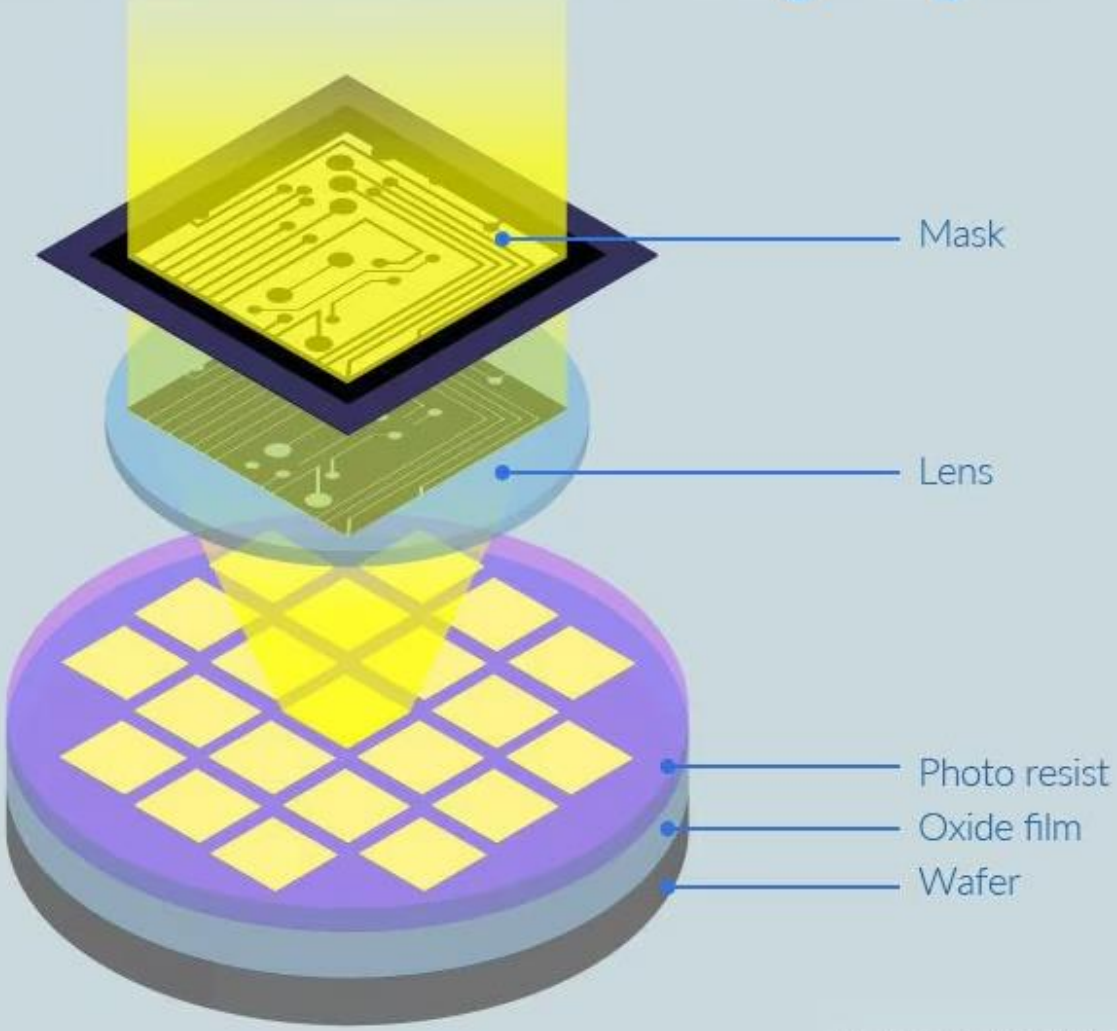





https://avesis.yildiz.edu.tr/ayten/dokumanlar

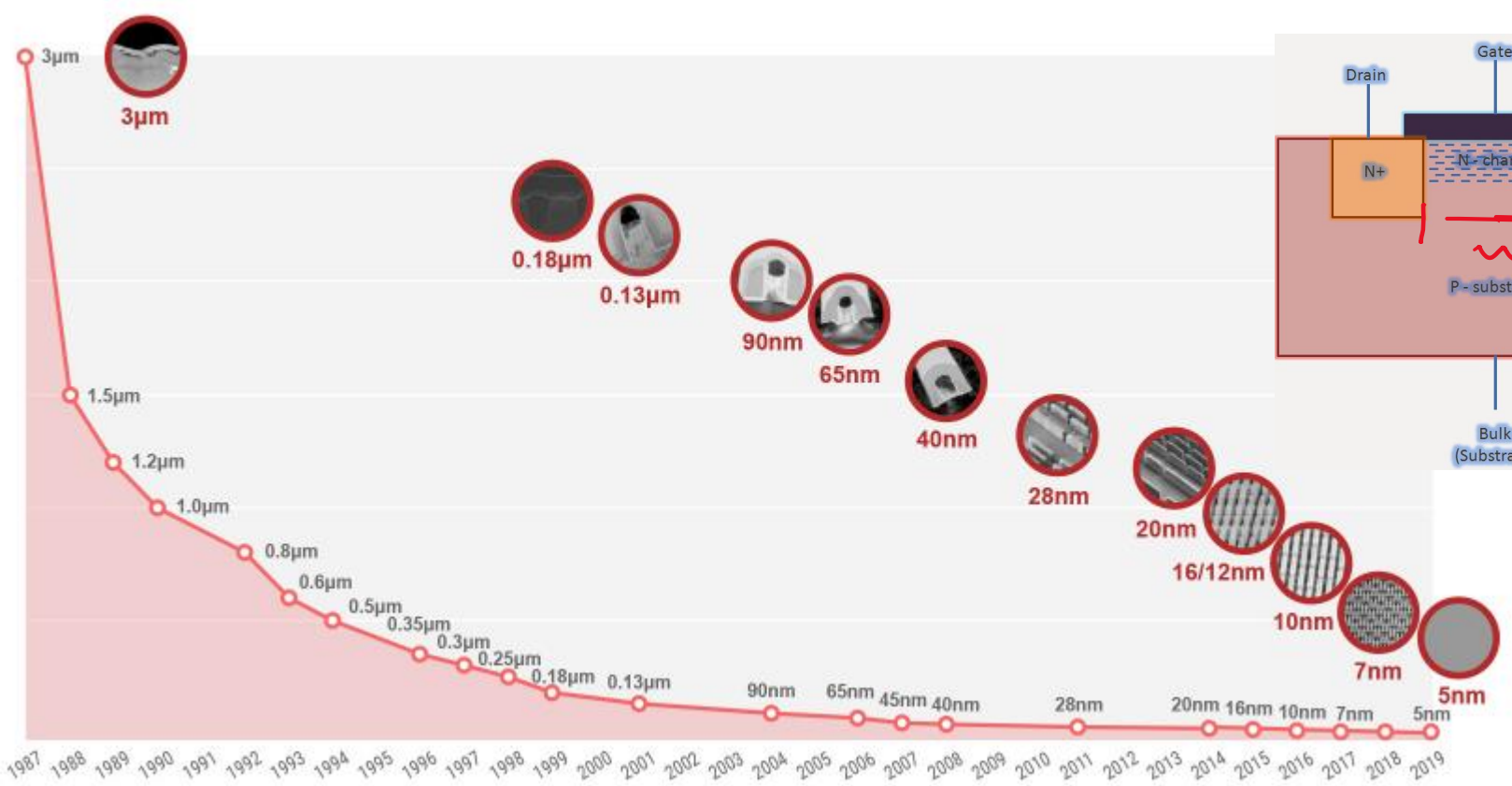
https://www.hardwaresecrets.com/how-chips-are-manufactured/

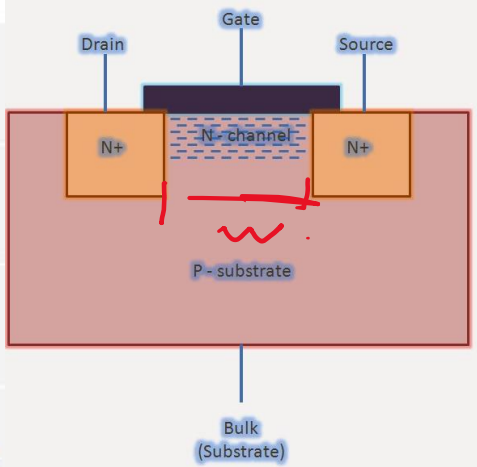
# Photolithography that draws circuits on the wafer through light

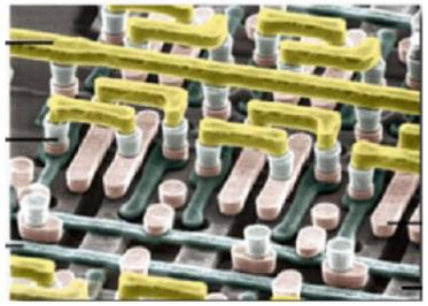


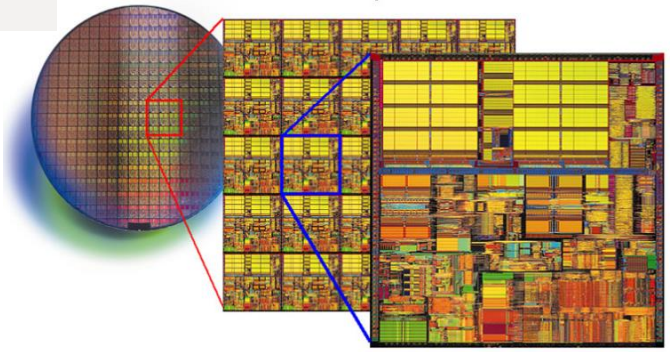

Samsung Semiconstory
samsungsemiconstory.com

# Mikroişlemcileri: Elektronik Devre Temelleri





https://www.tsmc.com/english/dedicatedFoundry/technology/logic/l_7nm

https://avesis.yildiz.edu.tr/ayten/dokumanlar

https://www.hardwaresecrets.com/how-chips-are-manufactured/

**Semiconductor device fabrication**

MOSFET scaling (process nodes)

- 10 µm – 1971
- 6 µm – 1974
- 3 µm – 1977
- 1.5 µm – 1981
- 1 µm – 1984
- 800 nm – 1987
- 600 nm – 1990
- 350 nm – 1993
- 250 nm – 1996
- 180 nm – 1999
- 130 nm – 2001
- 90 nm – 2003
- 65 nm – 2005
- 45 nm – 2007
- 32 nm – 2009
- 22 nm – 2012
- 14 nm – 2014
- 10 nm – 2016
- 7 nm – 2018
- 5 nm – 2020
- 3 nm – 2022

Future

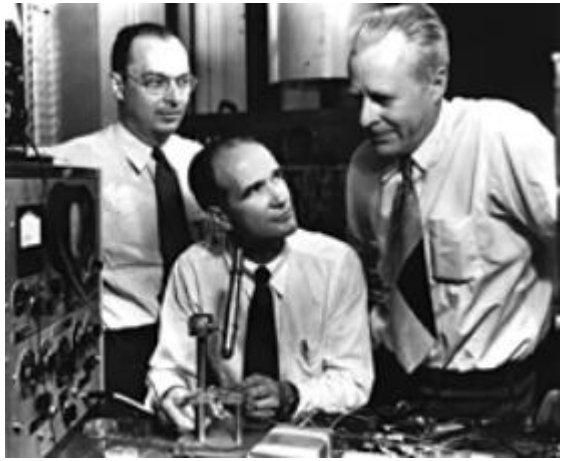
- 2 nm ~ 2024

| Processor | MOS transistor count | Date of introduction | Designer | MOS process (nm) | Area (mm$^2$) | Transistor density, tr./mm$^2$ |
|---|---|---|---|---|---|---|
| Apple M2 Pro (12-core 64-bit ARM64 SoC, SIMD, caches) | 40,000,000,000[177] | 2023 | Apple | 5 nm | ? | ? |
| Apple M2 Max (12-core 64-bit ARM64 SoC, SIMD, caches) | 67,000,000,000[177] | 2023 | Apple | 5 nm | ? | ? |

| Processor | Transistor count | Year | Designer(s) | Fab(s) | MOS process | Area | Transistor density, tr./mm$^2$ |
|---|---|---|---|---|---|---|---|
| AD102 Ada Lovelace | 76,300,000,000 | 2022 | Nvidia | TSMC | 4 nm | 608.4 mm² | 125,411,000 |
| AD103 Ada Lovelace | 45,900,000,000 | 2022 | Nvidia | TSMC | 4 nm | 378.6 mm² | 121,240,000 |
| AD104 Ada Lovelace | 35,800,000,000 | 2022 | Nvidia | TSMC | 4 nm | 294.5 mm² | 121,560,000 |
| Navi 31 RDNA3 | 58,000,000,000 | 2022 | AMD | TSMC | 5 nm (GCD)<br>6 nm (MCD) | 531 mm² (MCM)<br>306 mm² (GCD)<br>6×37.5 mm² (MCD) | 109,200,000 (MCM)<br>132,400,000 (GCD) |

https://en.wikipedia.org/wiki/Transistor_count

# Mikroişlemcileri: Elektronik Devre Temelleri



John Bardeen,
William Shockley
Walter Brattain
Bell Telephone Laboratories
Yıl: 1948

https://www.electronicsweekly.com/news/first-transistor-created-70-years-ago-the-device-that-changed-the-world-2017-12/

https://www.nature.com/articles/41012

# Mikroişlemcileri: Elektronik Devre Temelleri

https://hackaday.com/wp-content/uploads/2021/07/ena-computer-featured.jpg

# Sep 9, 1947 CE: World's First Computer Bug

On September 9, 1947, a team of computer scientists reported the world's first computer bug—a moth trapped in their computer at Harvard University.

**GRADES**
3 - 12

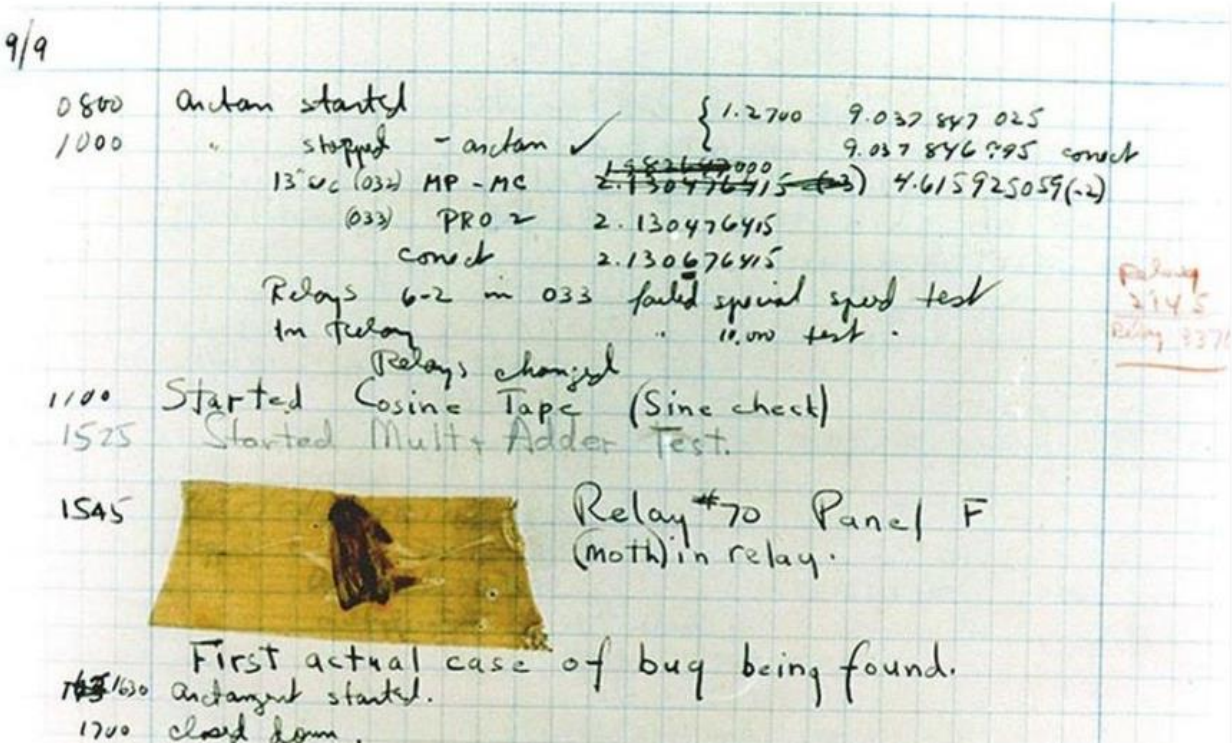
**SUBJECTS**
English Language Arts, Experiential Learning

PHOTOGRAPH

## Computer Bug

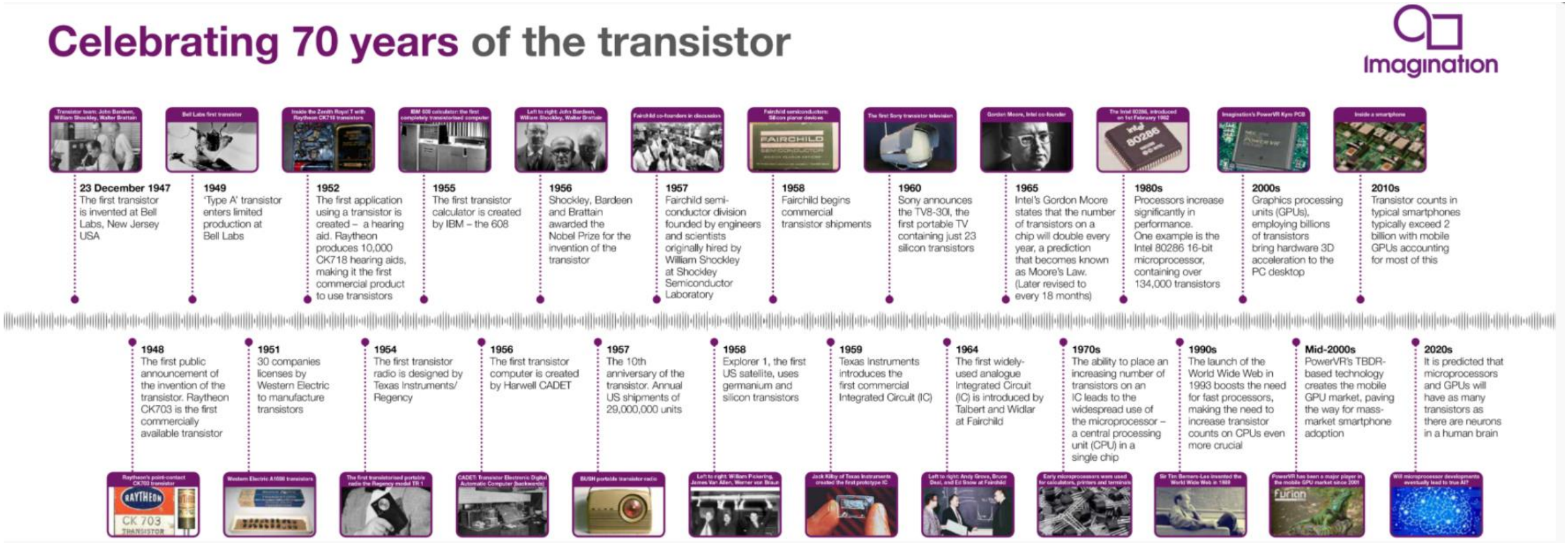
"First actual case of bug being found," according to the brainiacs at Harvard, 1945. The engineers who found the moth were the first to literally "debug" a machine.

> PHOTOGRAPH COURTESY NAVAL SURFACE WARFARE CENTER, DAHLGREN, VIRGINIA




https://education.nationalgeographic.org/resource/worlds-first-computer-bug/

# Mikroişlemcileri: Elektronik Devre Temelleri



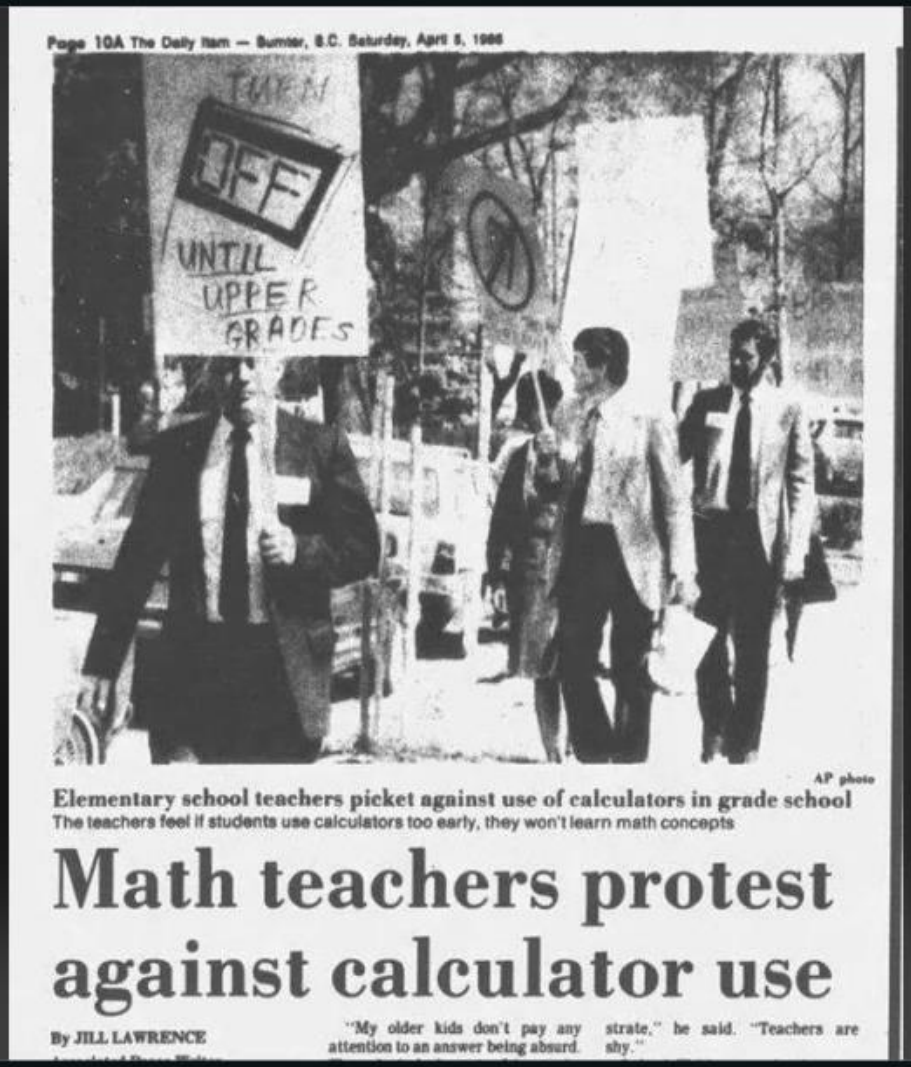

https://www.electronicsweekly.com/news/first-transistor-created-70-years-ago-the-device-that-changed-the-world-2017-12/

https://avesis.yildiz.edu.tr/ayten/dokumanlar

r/Damnthatsinteresting • 1 yr. ago
4reddityo

# Math teachers protest against calculator use 1966

Image



Page 10A The Daily Item — Sumter, S.C. Saturday, April 5, 1966



AP photo

**Elementary school teachers picket against use of calculators in grade school**

The teachers feel if students use calculators too early, they won't learn math concepts

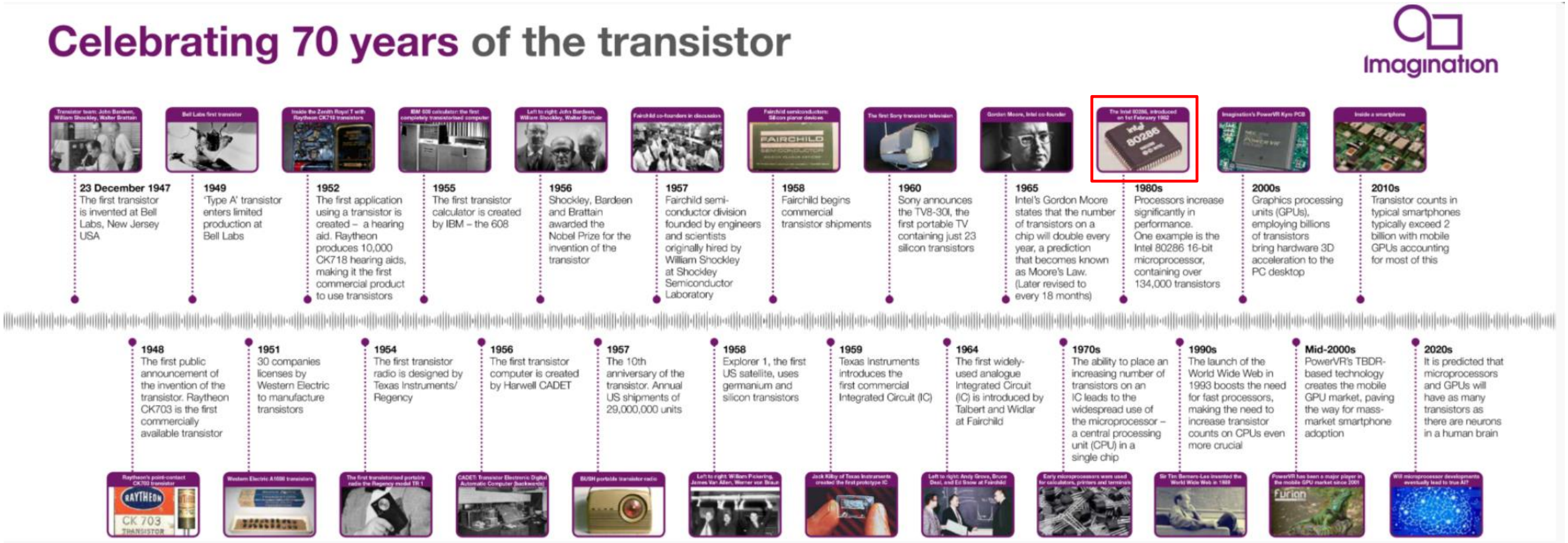
## Math teachers protest against calculator use

By JILL LAWRENCE

"My older kids don't pay any attention to an answer being absurd. strate," he said. "Teachers are shy."

https://www.reddit.com/r/Damnthatsinteresting/comments/11x0vyz/math_teachers_protest_against_calculator_use_1966/?rdt=41341

# Mikroişlemcileri: Elektronik Devre Temelleri

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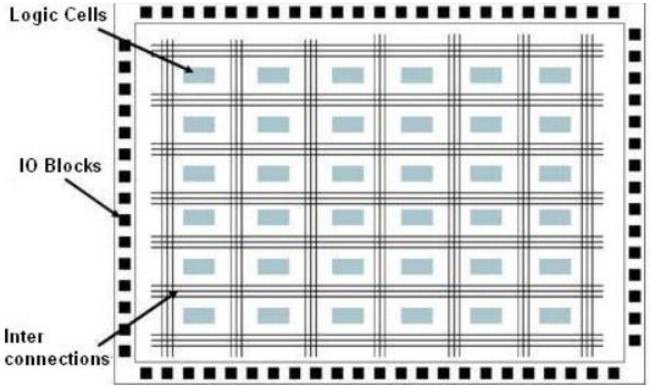
https://avesis.yildiz.edu.tr/ayten/dokumanlar

# Mikroişlemcileri: Elektronik Devre Temelleri

## Mikroişlemci

## Mikrodenetleyici

## FPGA



# Mikroişlemcileri: İkili Sayı Sistemi

İkili Sayı Sistemi

# Mikroişlemcileri: İkili Sayı Sistemi

## İkili Sayı Sistemi

**George Boole**

The Mathematical Analysis of Logic (1847)

Boole Cebri ile üç temel lojik işlem:

VE (AND), VEYA (OR) ve TÜMLEME (DEĞİL-NOT)

Teorik gelişmeler

pratik çalışmalardan genellikle 50 yıl ileridedir. !!

(Hatta 100 yıl)

# Mikroişlemcileri: İkili Sayı Sistemi

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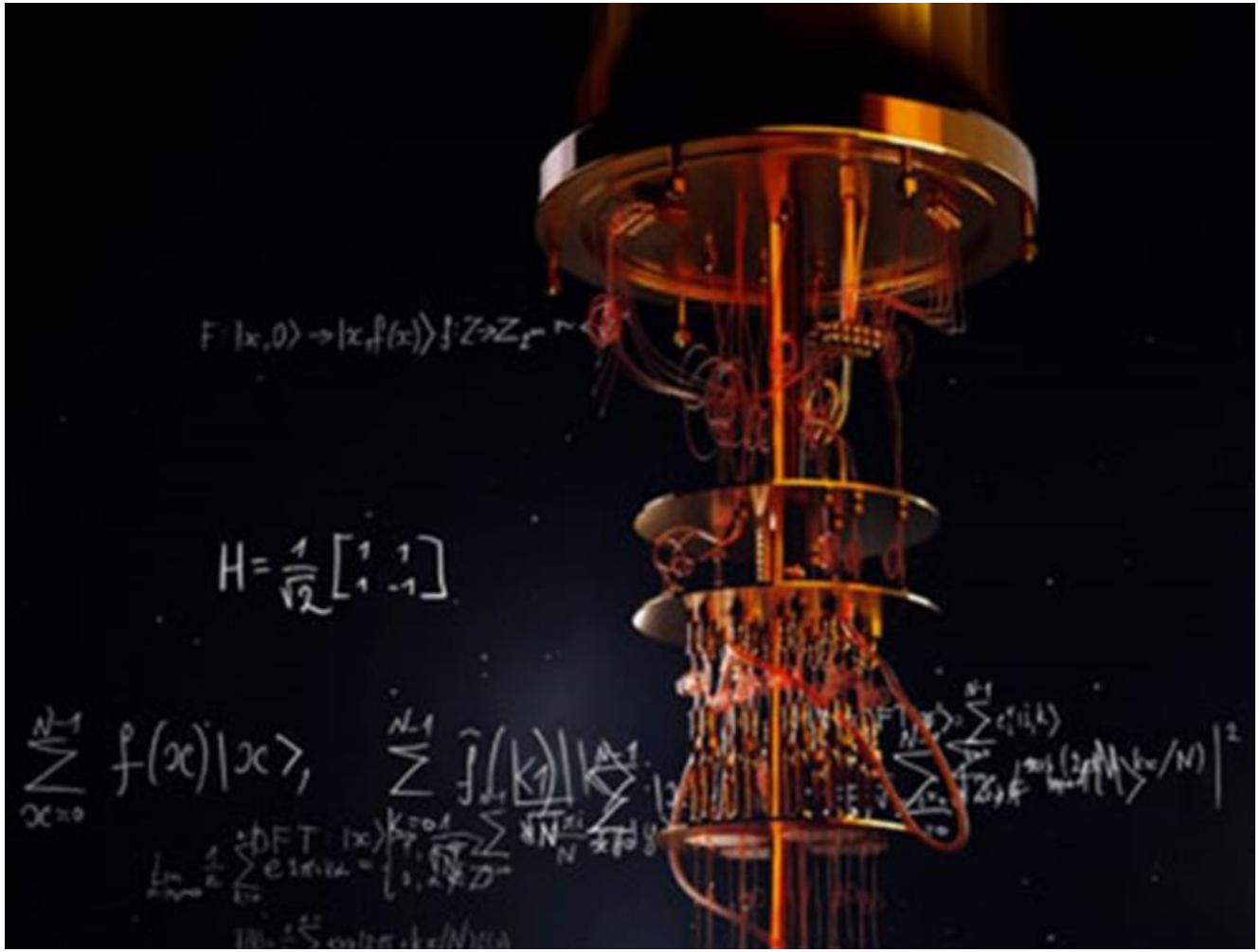

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Teorik gelişmeler

pratik çalışmalardan genellikle 50 yıl ileridedir. !!

(Hatta 100 yıl)

**ve (and) kanunu**

A.1=A

A.0=0

**veya (or) kanunu**

A+1=1

A+0=A

**Yutma Kuralı**

A.(A+B)=A

A + AB =A

**Dağılma Kuralı** [c

A(B+C)=AB+AC

A+B.C=(A+B)(A+C)

**De Morgan Kuralı**

(A.B)'=A'+B'

(A+B)'=A'.B'

# Mikroişlemcileri: İkili Sayı Sistemi

- Sayı Sistemleri: 10 , 2 , 16

**Table 1: Binary Information Group Representations and Terms**

| Number of Bits | Common Representation Terms |
|---|---|
| 1 | Bit / Digit / Flag |
| 4 | Nybble / Nibble |
| 8 | Byte / Octet / Character |
| 16 | Double Byte / Word |
| 32 | Double Word / Long Word |
| 64 | Very Long Word |

http://www.tcpipguide.com/free/t_BinaryInformationandRepresentationBitsBytesNibbles-2.htm

2'lik Sayı Sistemi

Sürekli 2'ye bölünür kalanlar bütünleştirilir.

16'lık Sayı Sistemi

| | | | |
|---|---|---|---|
| 125 | 16 | 7 | 13 |
| − 112 | 7 | + 7 | $D$ |
| 13 | | | |

| | | | | |
|---|---|---|---|---|
| Tam | 12345 | 771 | 48 | 3 |
| Kalan | 9 | 3 | 0 | |

Hex 3 0 3 9

```
>> dec2hex(12345)

ans =

    '3039'
```

# Mikroişlemcileri: İkili Sayı Sistemi

2'lik Sayı Sisteminde İşlemler
(İşaretsiz)

| 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1 | |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 185 |

| 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1 | |
|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 27 |

# Mikroişlemcileri: İkili Sayı Sistemi

2'lik Sayı Sisteminde İşlemler
(İşaretsiz)

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 185 |
| 0 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 27 |
| 1 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 212 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | E(C) |

```
>> dec2bin(212)

ans =

    '11010100'
```

# Mikroişlemcileri: İkili Sayı Sistemi

2'lik Sayı Sisteminde İşlemler
(İşaretsiz)

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 185 |
| 0 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 27 |
| 1 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 212 |
| 0 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | E(C) |

```
>> dec2bin(212)

ans =

    '11010100'
```

2'lik Sayı Sisteminde İşlemler
(İşaretli)

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 27 |
| 1 | 1 | 1 | 0 | 0 | 1 | 0 | 1 | -27 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | E(C) |

# Mikroişlemcileri: İkili Sayı Sistemi

2'lik Sayı Sisteminde İşlemler
(İşaretsiz)

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 185 |
| 0 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 27 |
| 1 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 212 |
| 0 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | E(C) |

```
>> dec2bin(212)

ans =

    '11010100'
```

2'lik Sayı Sisteminde İşlemler
(İşaretli)

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 27 |
| 1 | 1 | 1 | 0 | 0 | 1 | 0 | 1 | -27 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | E(C) |

1

# Mikroişlemcileri: İkili Sayı Sistemi

2'lik Sayı Sisteminde İşlemler
(İşaretsiz)

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 185 |
| 0 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 27 |
| 1 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 212 |
| 0 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | E(C) |

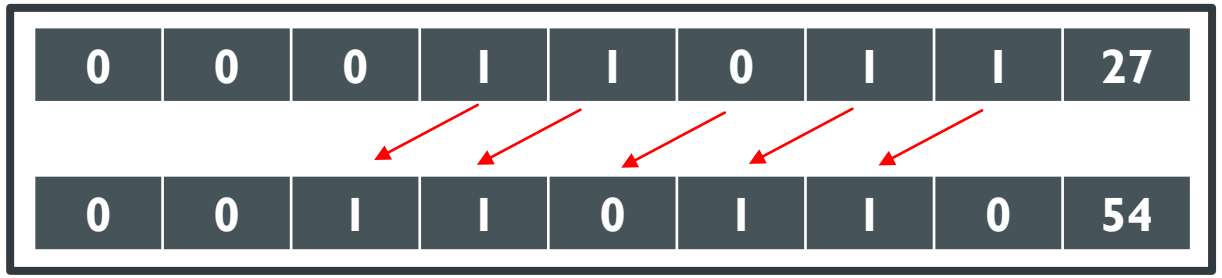
```
>> dec2bin(212)

ans =

    '11010100'
```

2'lik Sayı Sisteminde İşlemler
(İşaretli)

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 108 |
| 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | -108 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | E(C) |

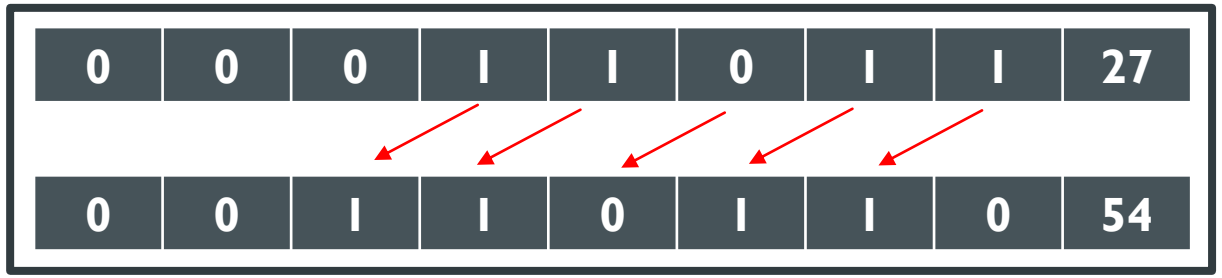
# Mikroişlemcileri: İkili Sayı Sistemi

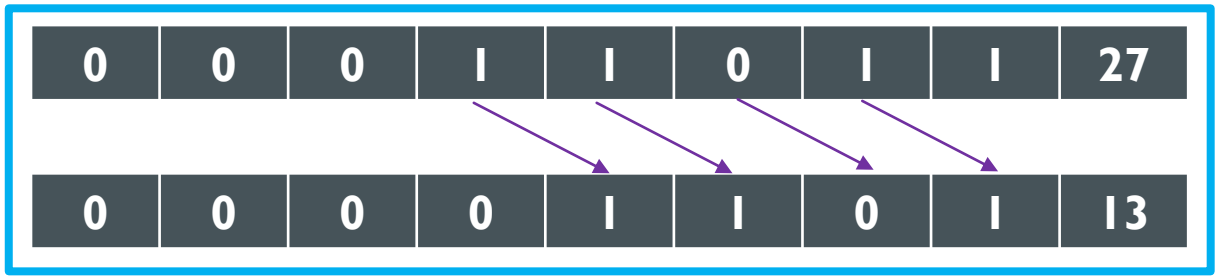
Çarpma/Bölme İşlemi



# Mikroişlemcileri: İkili Sayı Sistemi

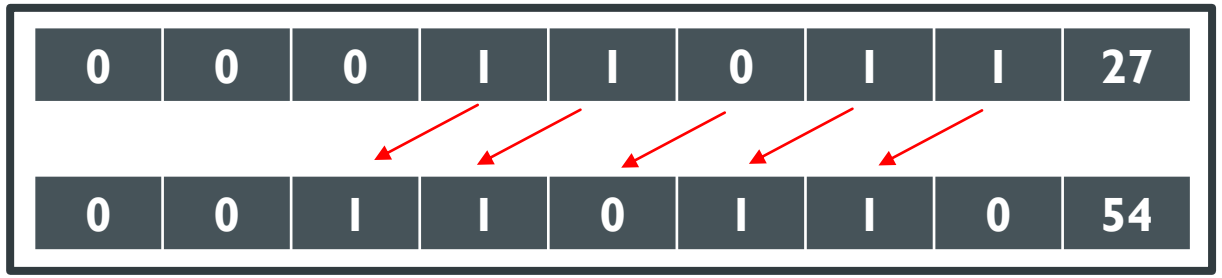
Çarpma/Bölme İşlemi

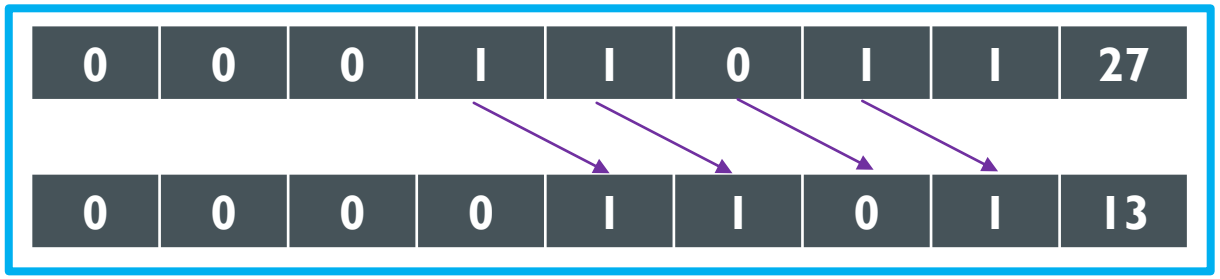


# Mikroişlemcileri: İkili Sayı Sistemi

Çarpma/Bölme İşlemi



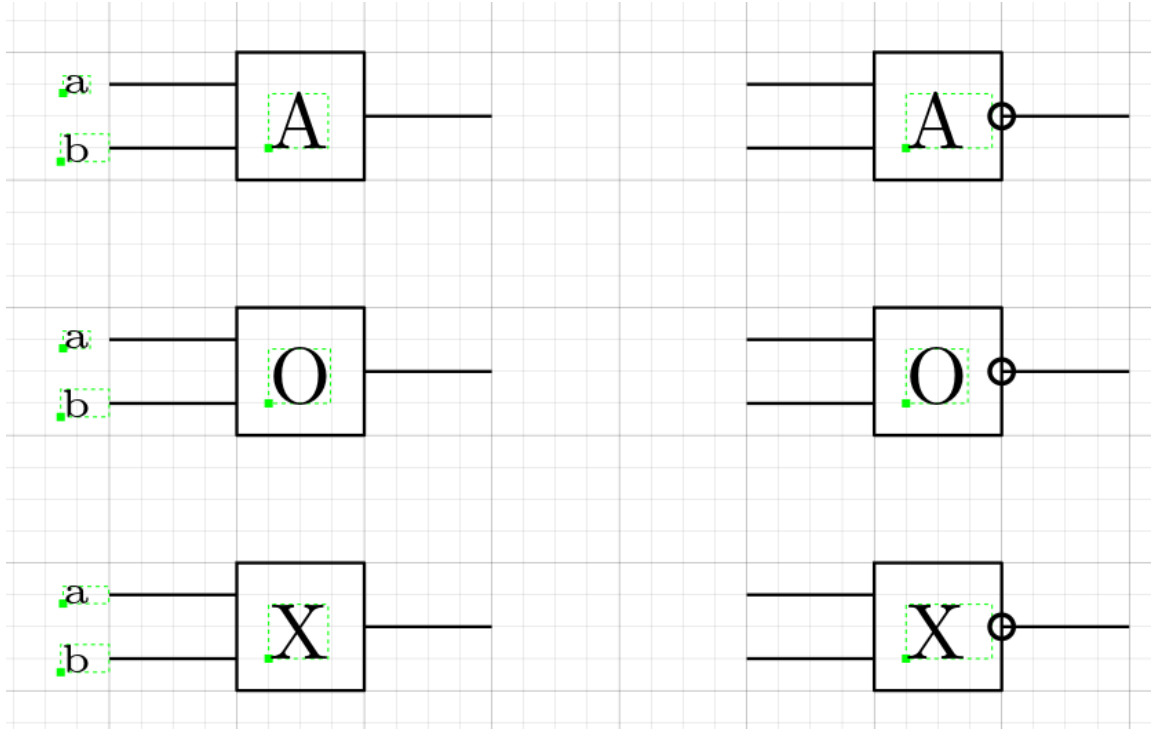




# 3 ile çarpma ?

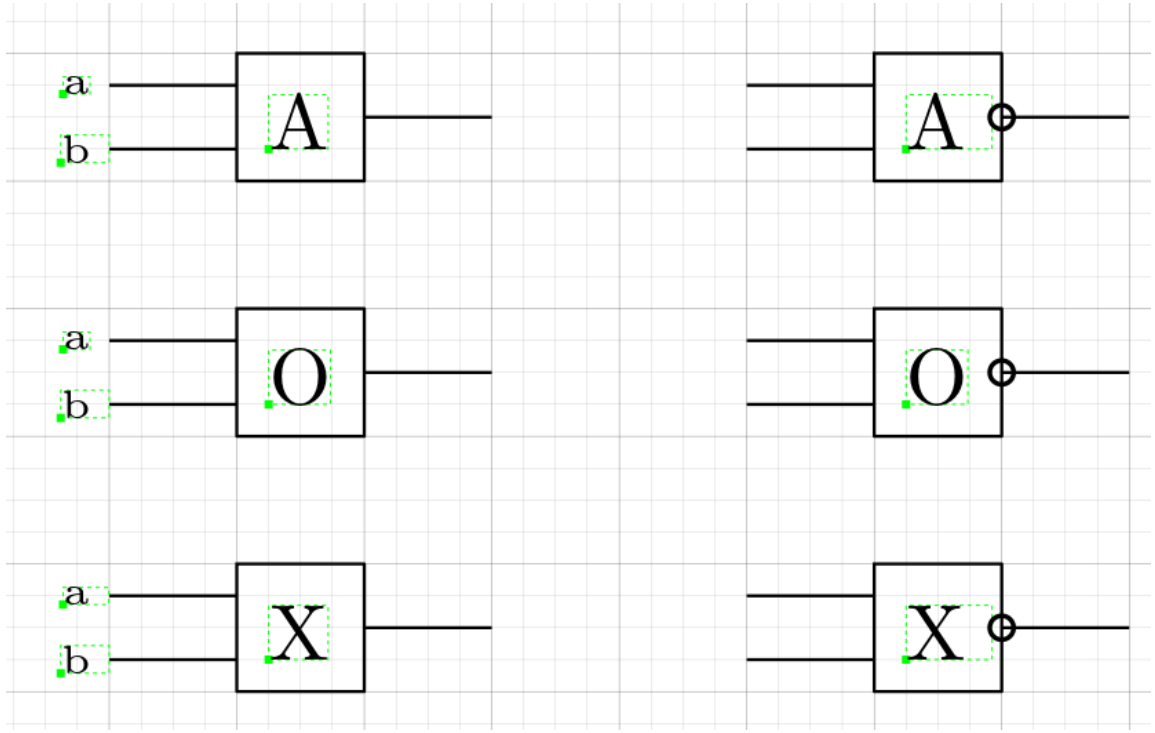
# LOJİK DEVRE TEMELLERİ

And, Or, XOR, MUX, Encoder

# Lojik Devreler: and , or , xor



# Lojik Devreler: and , or , xor

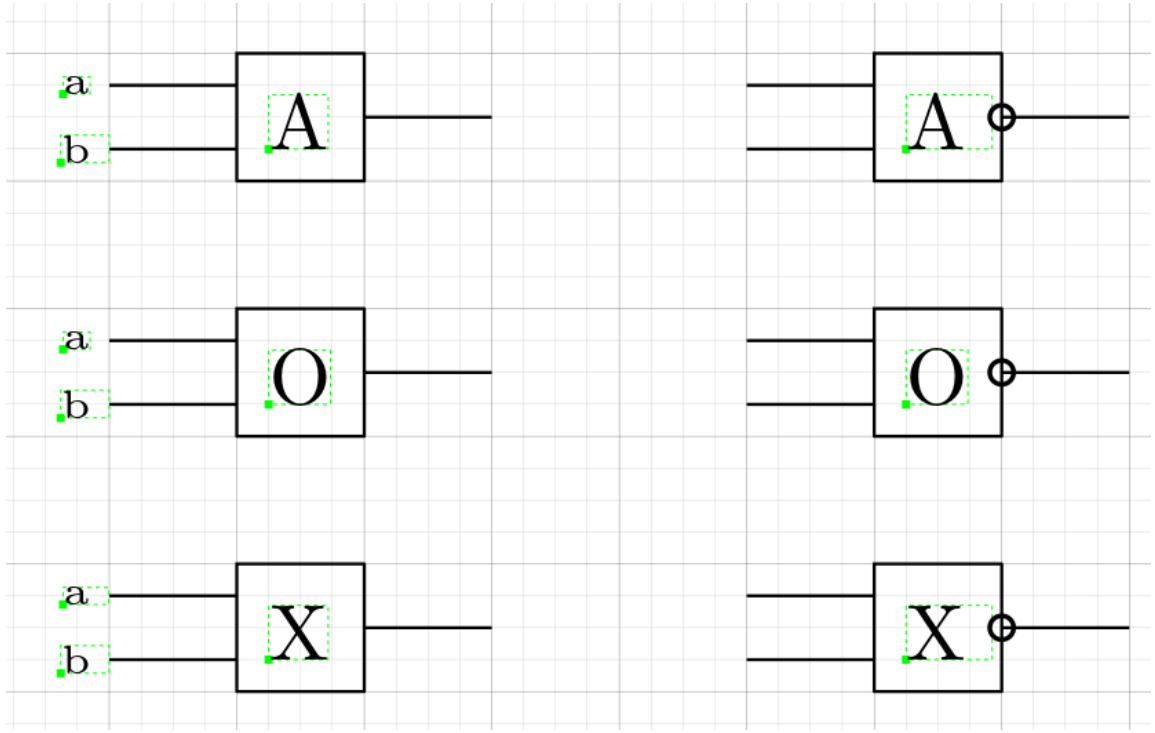



| a | b | ç |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

| a | b | ç |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

| a | b | ç |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

# Lojik Devreler: and , or , xor




| a | b | ç |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

AND

| a | b | ç |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

XOR

| a | b | ç |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

OR

# Lojik Devreler: Karnaugh Diyagramı

| a | b | c | Ç |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 0 |

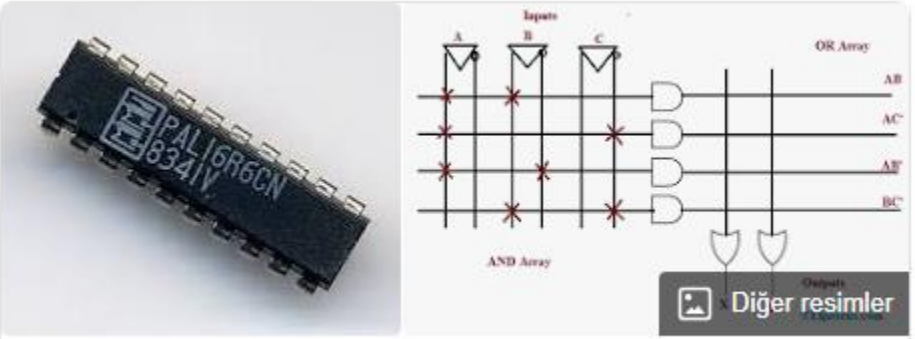
| ab \ c | 0 | 1 |
|---|---|---|
| 00 | 1 | 1 |
| 01 | 1 | 1 |
| 11 | 0 | 0 |
| 10 | 0 | 1 |

| ab \ c | 0 | 1 |
|---|---|---|
| 00 | 1 | 1 |
| 01 | 1 | 1 |
| 11 | 0 | 0 |
| 10 | 0 | 1 |

*minterm*

$$f = a' + b'c$$

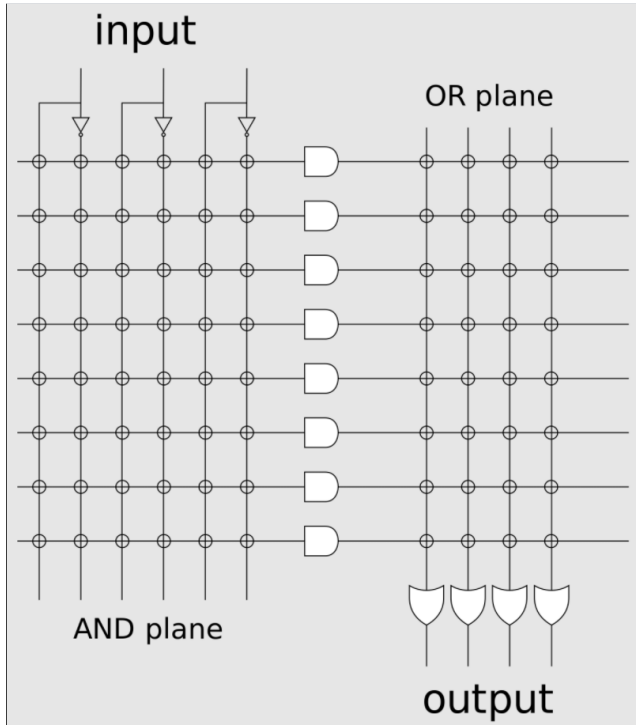
# Lojik Devreler: Karnaugh Diyagramı

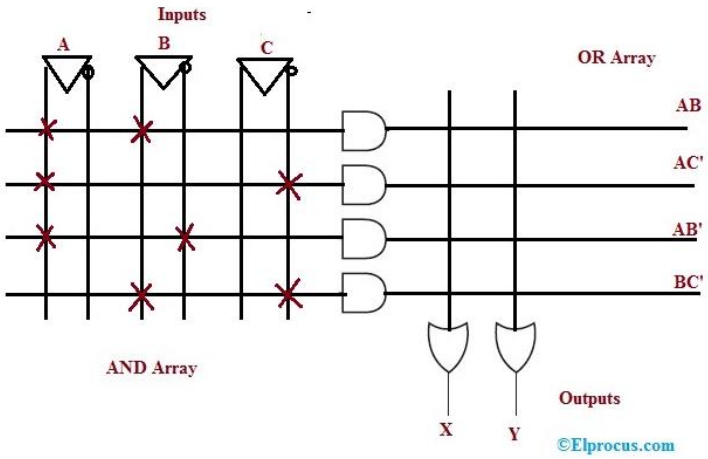



## Programmable Array Logic

İngilizceden çevrilmiştir - Programlanabilir Dizi Mantığı, Monolithic Memories, Inc. tarafından Mart 1978'de tanıtılan dijital devrelerde mantık işlevlerini uygulamak için kullanılan programlanabilir mantık aygıtı yarı iletkenlerinin bir ailesidir. Wikipedia (İngilizce)

Orijinal açıklamayı göster




https://commons.wikimedia.org/wiki/File:Programmable_logic_array_(schematic_drawing_example).svg

# Lojik Devreler: Yarı ve Tam Toplayıcı

- **Matematiğe Giriş**

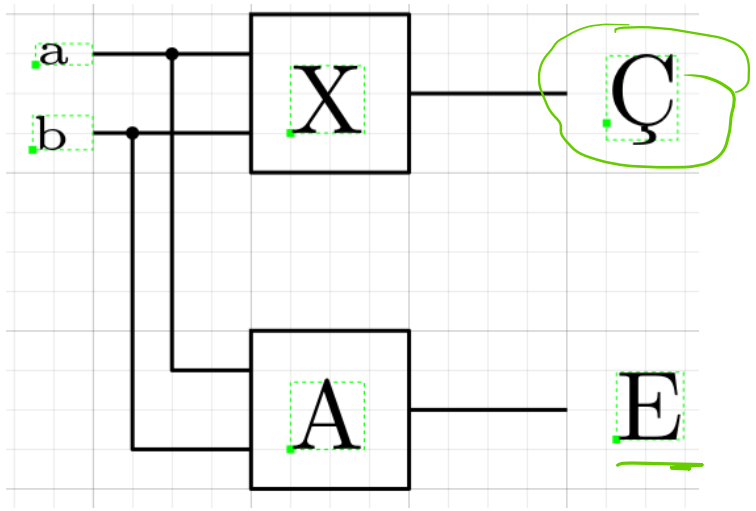
| a | b | Ç |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

# Lojik Devreler: Yarı ve Tam Toplayıcı

- **Matematiğe Giriş**

| a | b | Ç |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

1



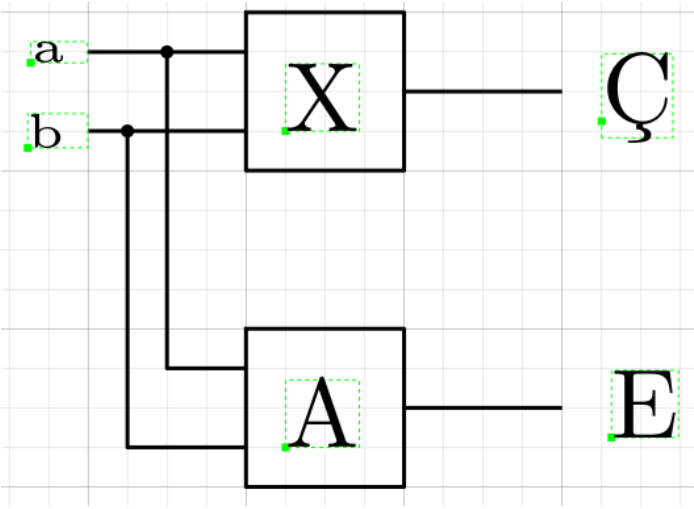

YARI TOPLAYICI

# Lojik Devreler: Yarı ve Tam Toplayıcı

- **Matematiğe Giriş**

| a | b | Ç |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

1




YARI TOPLAYICI

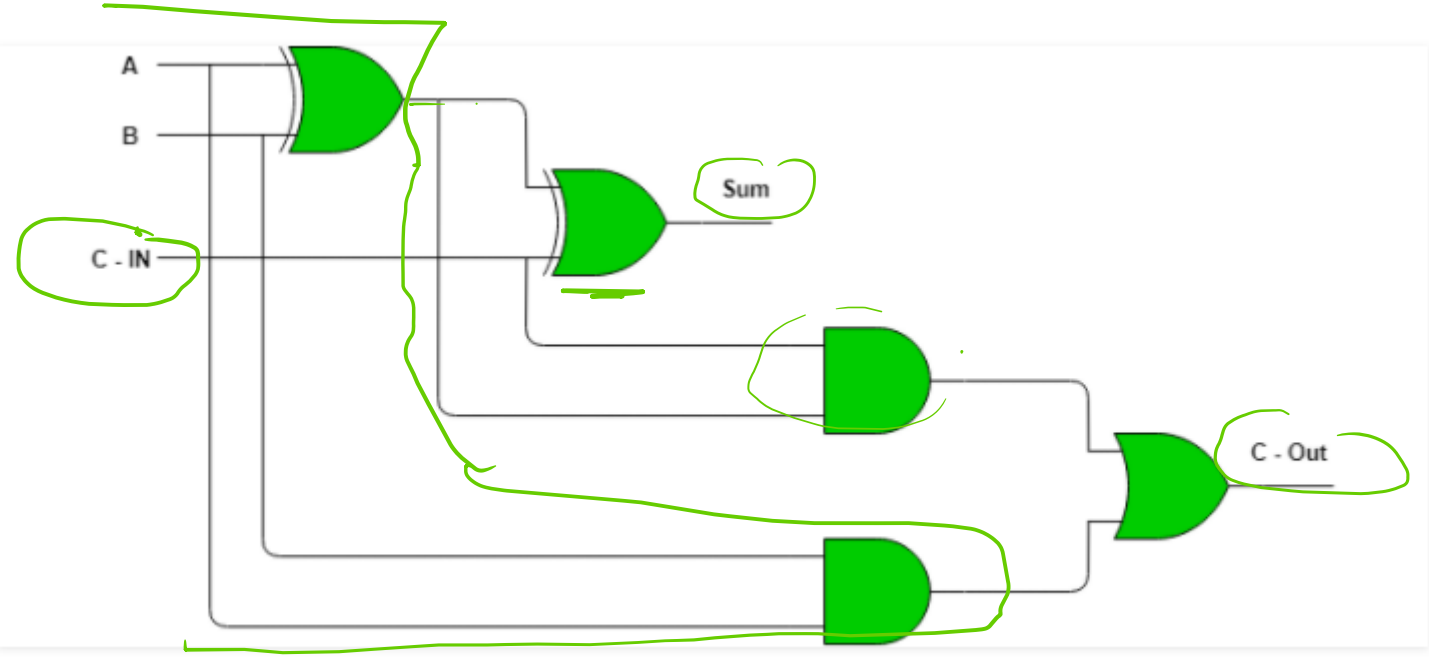
Elde Çıkışı Tamam

Elde Girişi ??

# Lojik Devreler: Yarı ve Tam Toplayıcı

- **Matematiğe Giriş**

| a | b | c | Ç | E |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 |



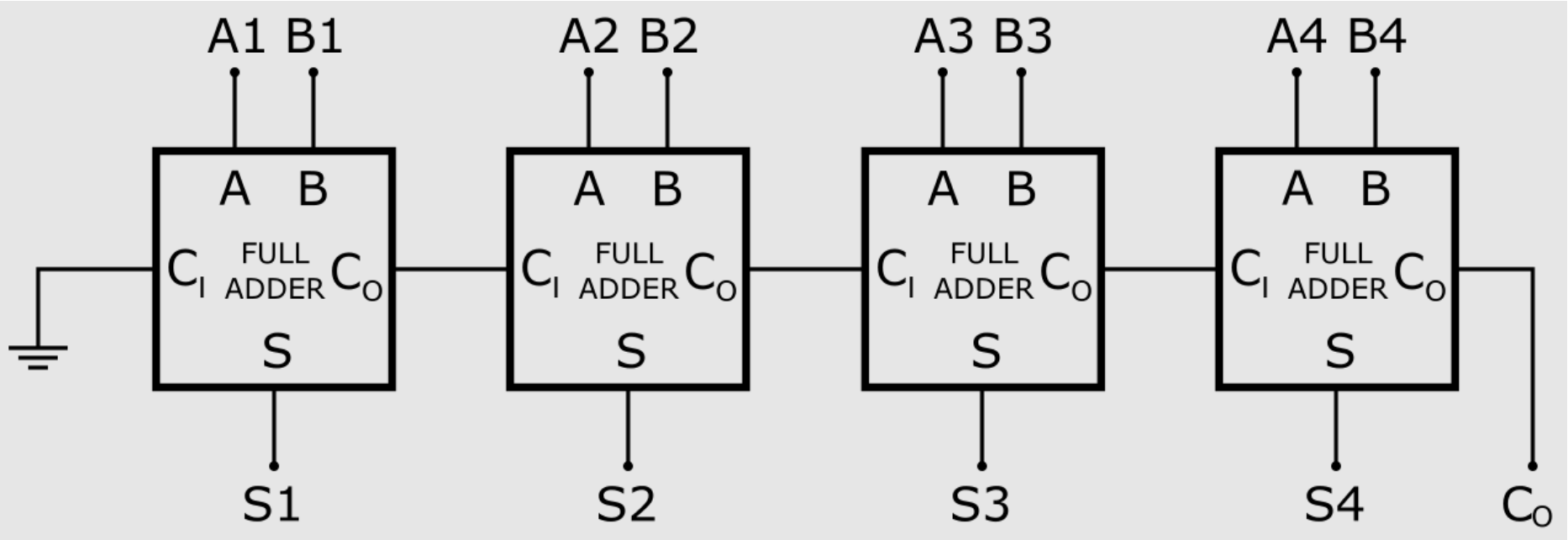

Full Adder logic circuit.

https://www.geeksforgeeks.org/full-adder-in-digital-logic/

# Lojik Devreler: Yarı ve Tam Toplayıcı

- **Matematiğe Giriş**

| a | b | c | Ç | E |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 |




https://upload.wikimedia.org/wikipedia/commons/thumb/f/fe/4-Bit_Full_Adder.svg/1280px-4-Bit_Full_Adder.svg.png

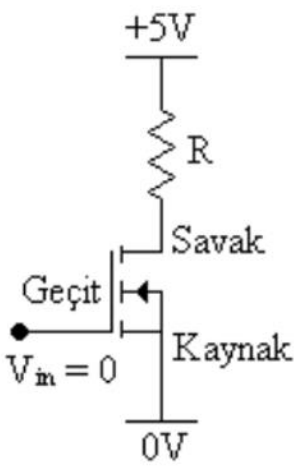
# Lojik Devreler: Yarı ve Tam Toplayıcı

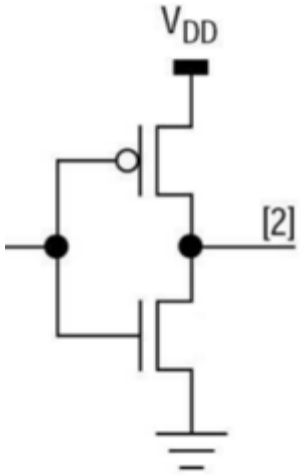
Transistor

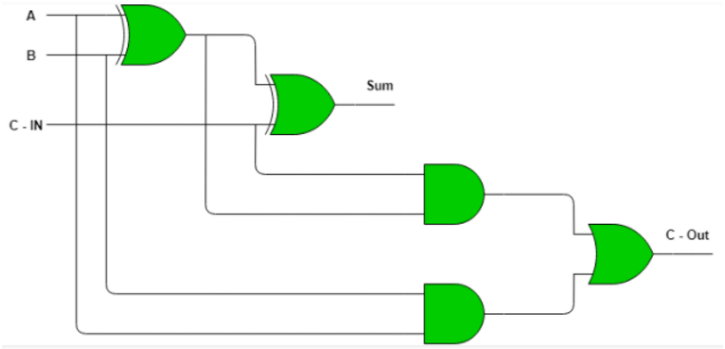
Kapılar

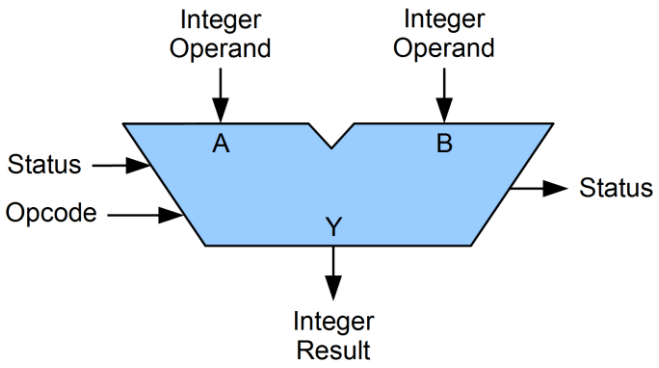
Hesaplayıcılar
Tam Toplayıcı , Çarpa , Bölme

Aritmatic Logic Unit (ALU)

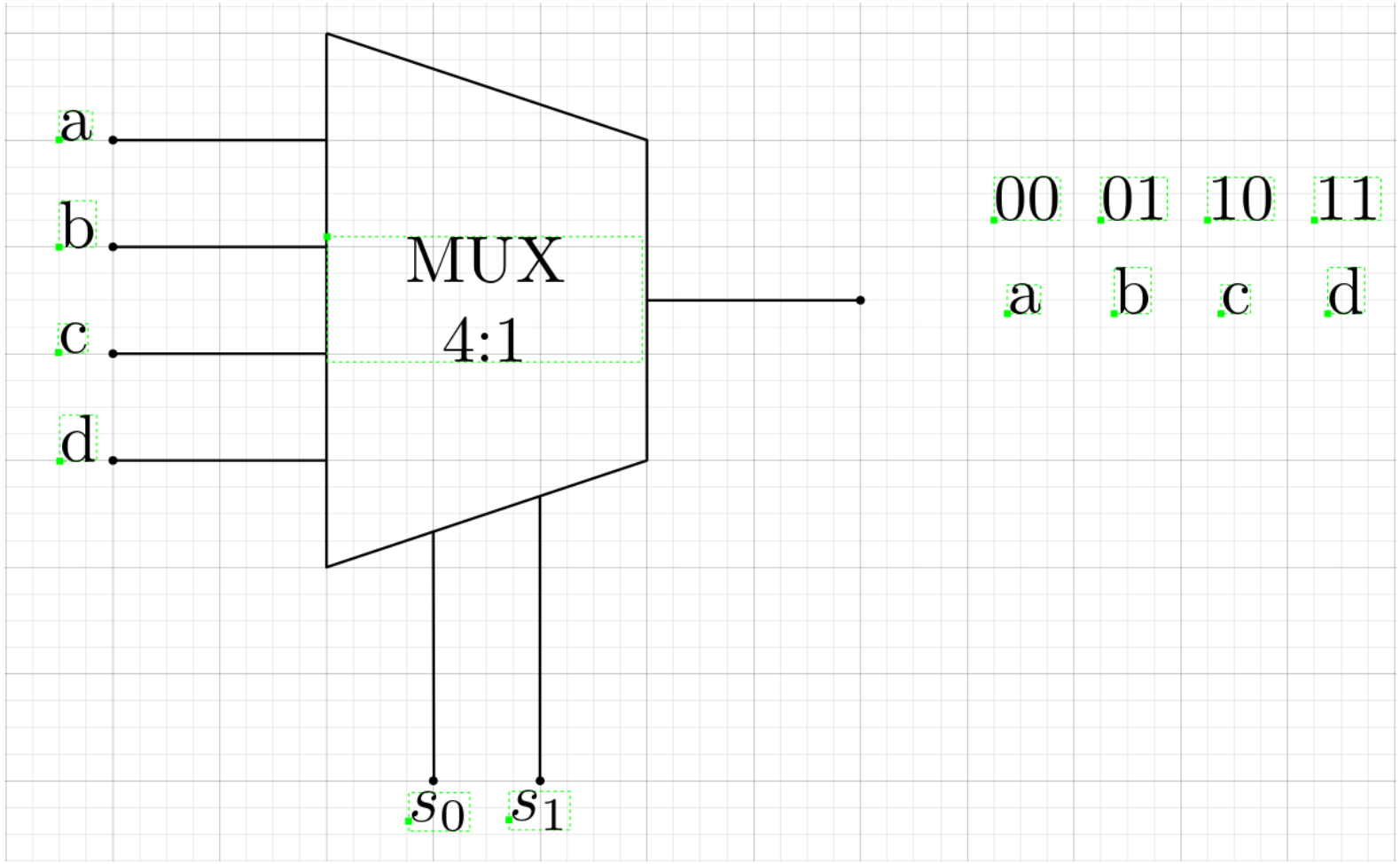


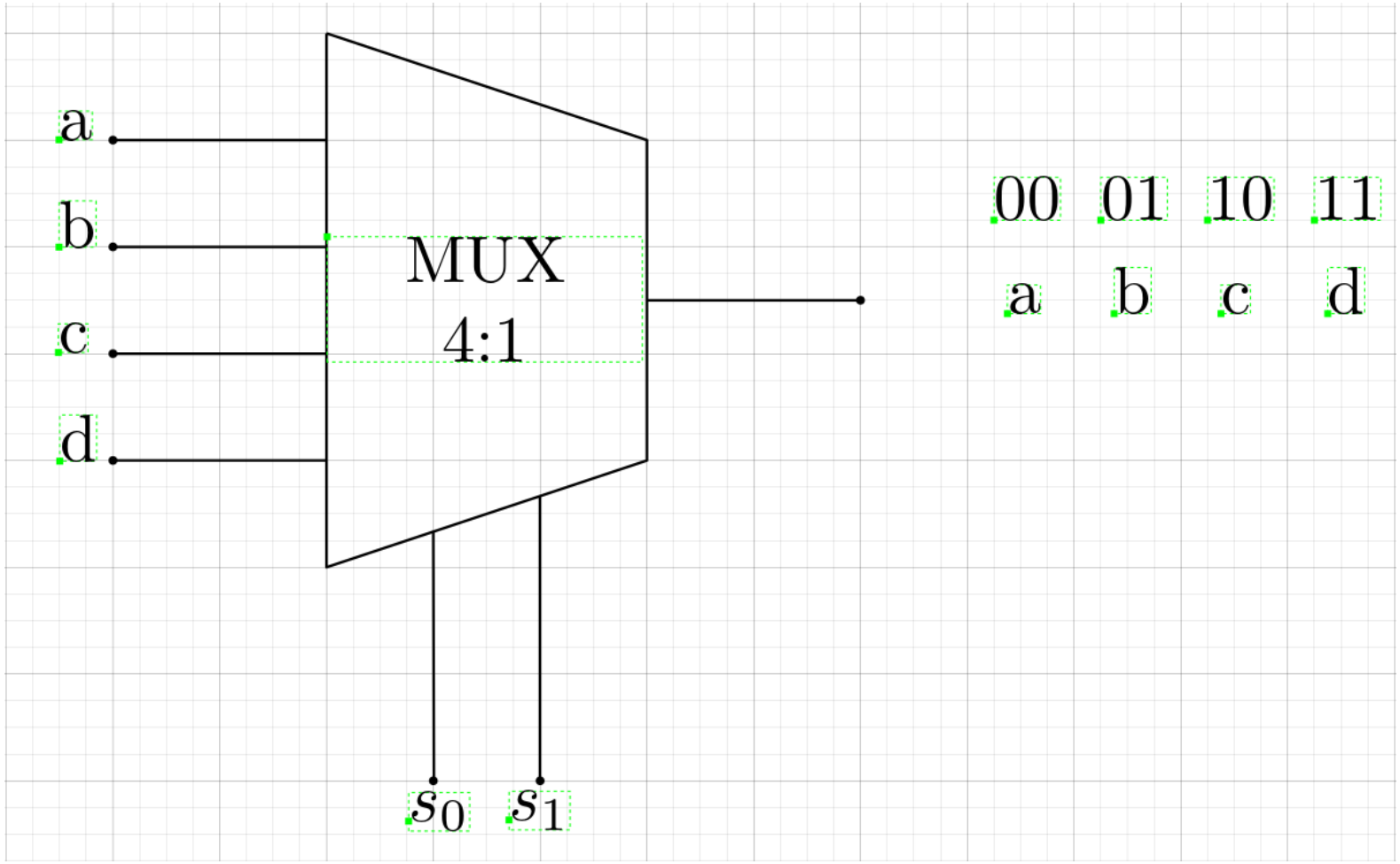



# Lojik Devreler: Çoğullayıcı (Multiplexer (MUX))




| 00 | 01 | 10 | 11 |
|---|---|---|---|
| a | b | c | d |

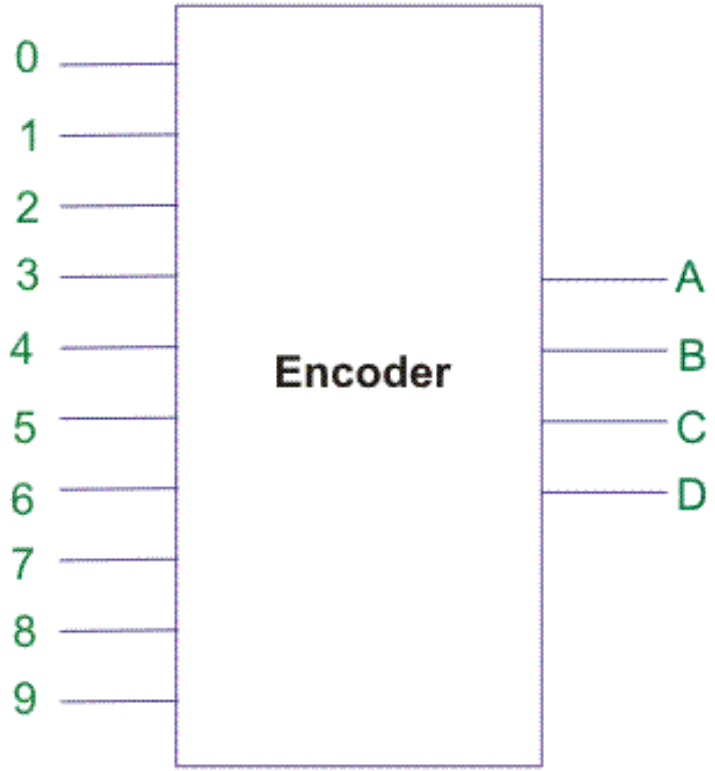
# Lojik Devreler: Çoğullayıcı (Multiplexer (MUX))




Paralel veriyi seri hale getirir

# Lojik Devreler: Kodlayıcı (BCD Encoder)

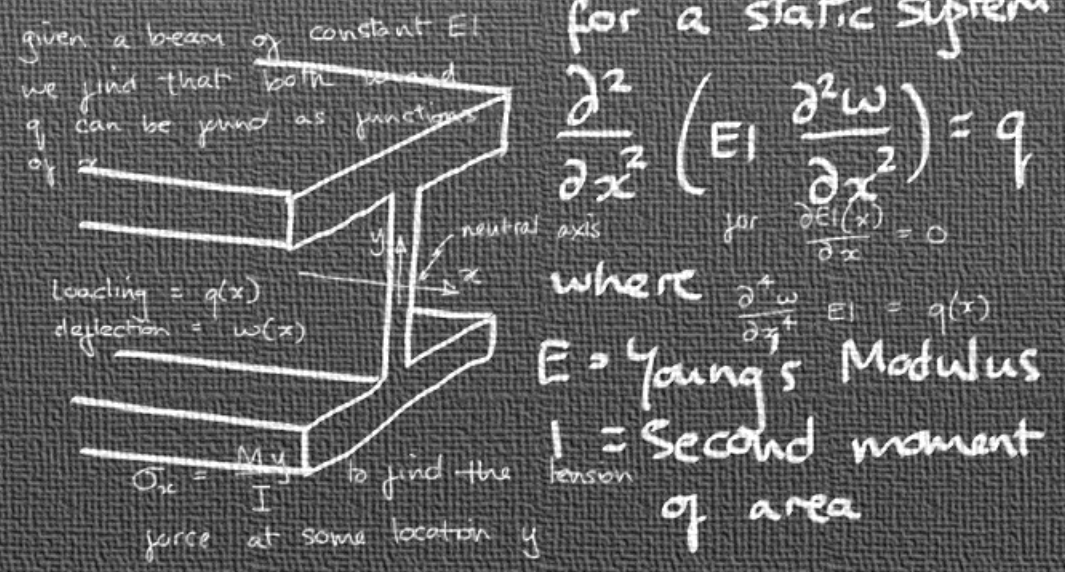
DEC BCD



# HAFTAYA GÖRÜŞMEK ÜZERE

## ARAŞTIRMA KONUSU:

Flip Flop Tutucu Yazmaç

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